

Machine Learning Principles

Class 22: November 24
Variational Auto Encoder (VAE)

Instructor: Diana Kim

- Clustering

[1] Clustering (Definition)

❖ **Clustering** : $x_n \in X \rightarrow k \in K = \{0, 1, 2, \dots, K - 1\}$

:partitioning data into a finite set of groups

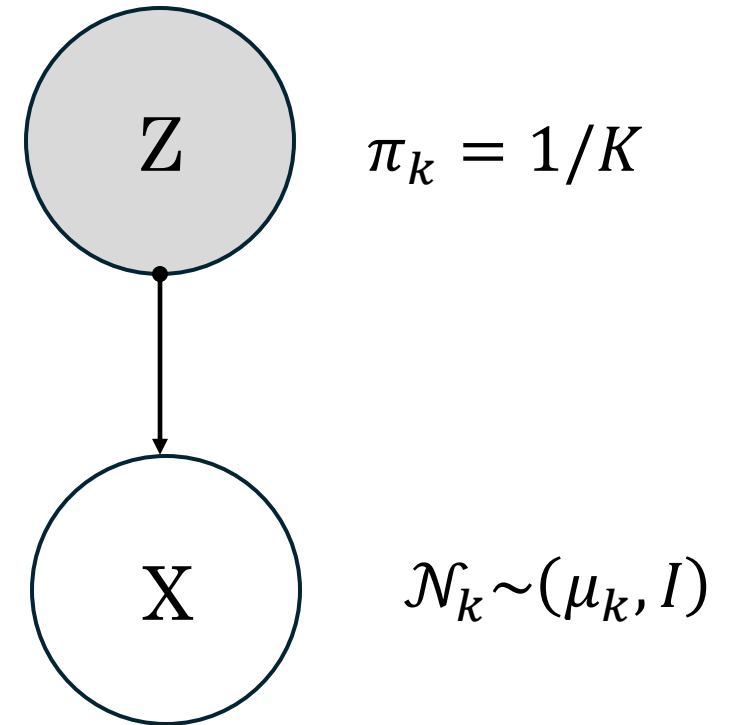
: within a group the data points are similar (minimize intra class variation)
or connected

: within a group the data points are dissimilar (maximize inter class variation)
or disconnected.

[2] Clustering (K-means)

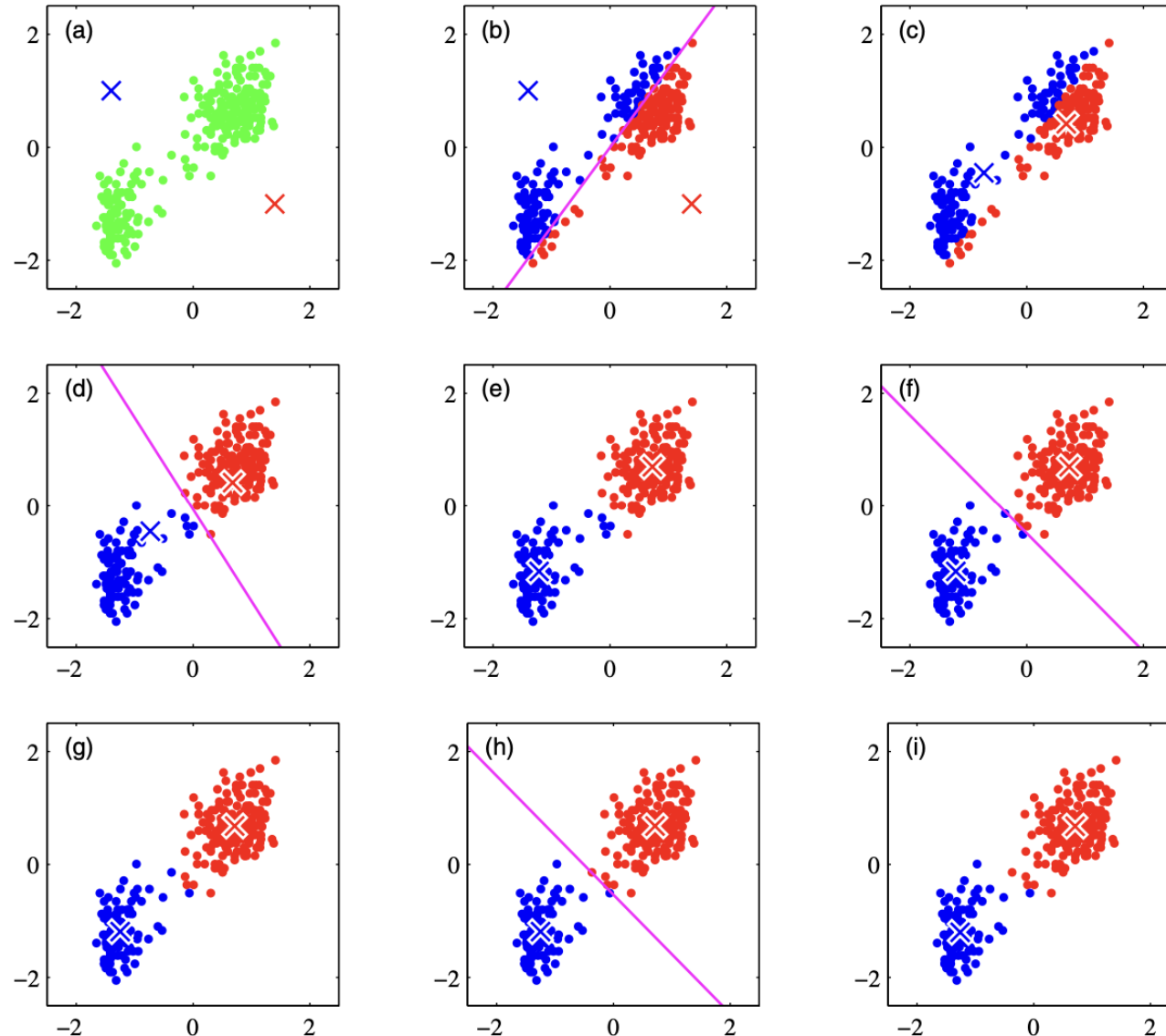
K-means algorithm

It fits the latent model with data.



Q: What would the shape of the clustering regions be?

[**] K-means Clustering (illustration of K-means progression)

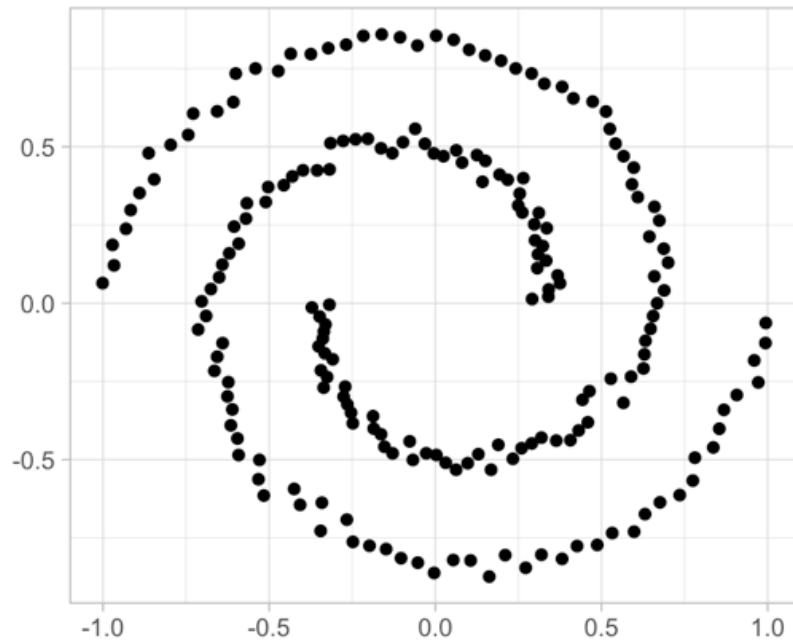


**continue until
there is no further changes
in the assignment.

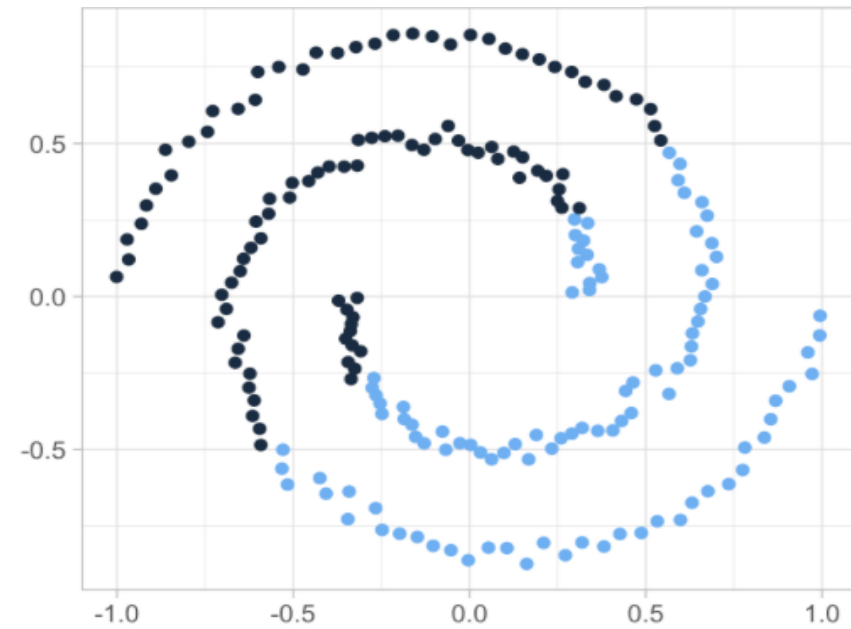
from textbook Bishop Fig. 9.1

[**] K-means Clustering (illustration of K-means progression)

(a) Original data



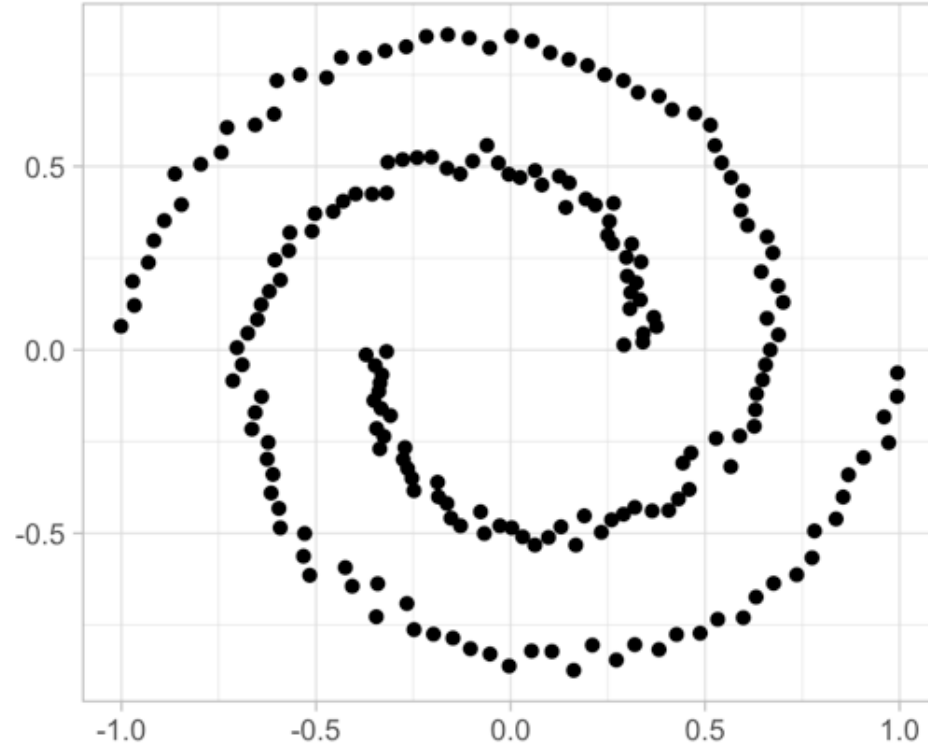
(b) K-means Cluster



Q: what if we define a cluster by a group of data points are densely packed.
but separated by sparse region from other clusters?

<https://bradleyboehmke.github.io/HOML/kmeans.html>

[**] K-means Clustering (illustration of K-means progression)



Q: How can we detect connected manifolds?
can you imagine a density of the data?

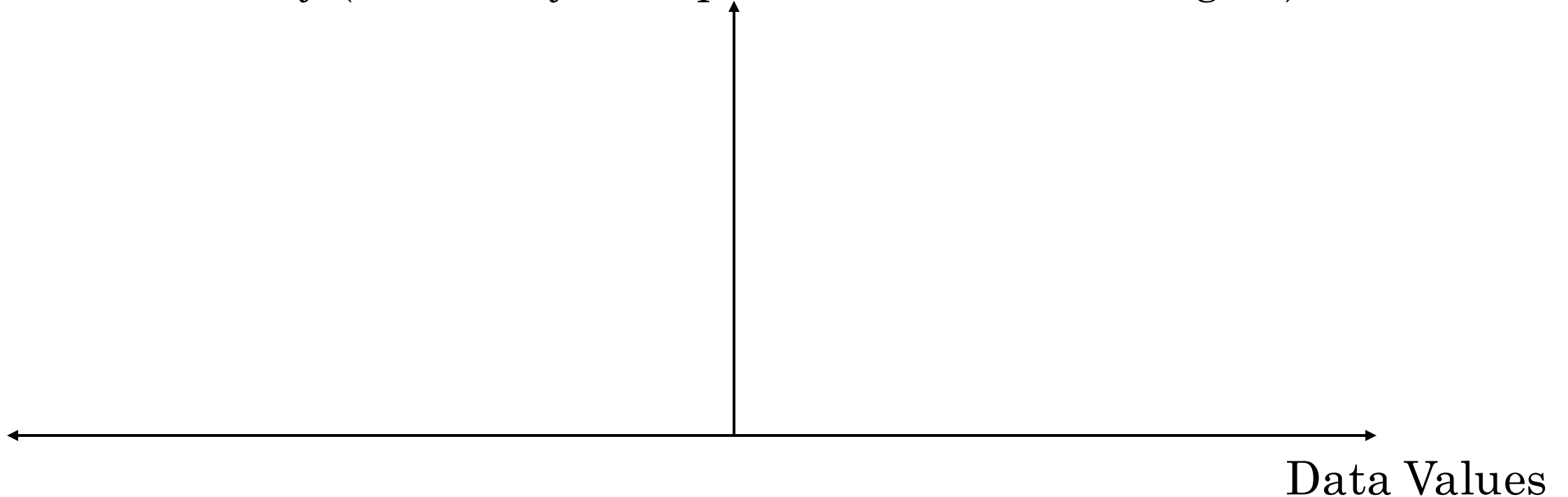
❖ Density Based Spatial Clustering

DBSCAN (Density Based Spatial Clustering of Application with Noise)

HDBSCAB (Hierarchical DBSCAN)

[1] Density Based Clustering (analogy of finding island)

density (how many data points exist within a region)



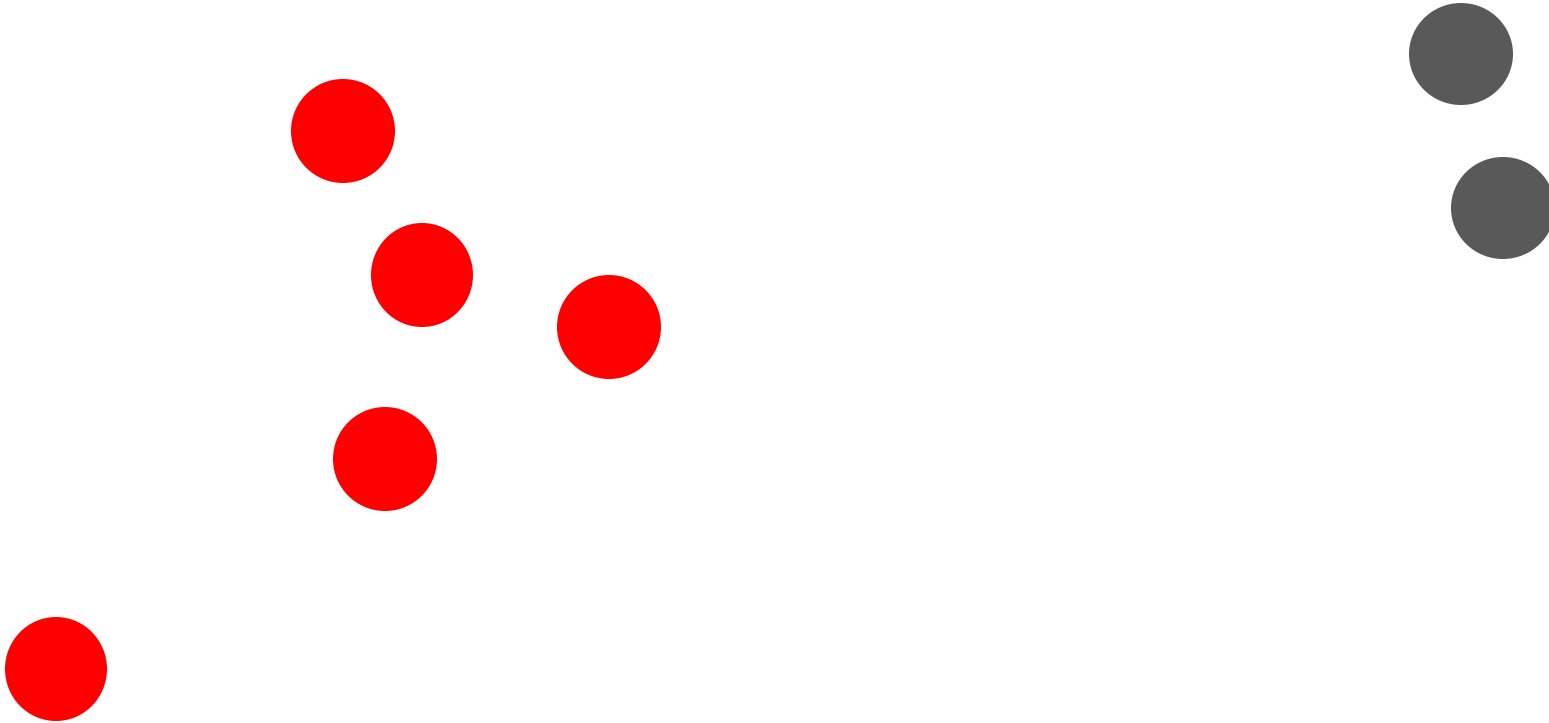
- island, emerging above the water:
- underwater land:
- water level:

- DBSCAN, non-parametric & density based
(Density Based Spatial Clustering of Application with Noise)

[paper review]

<https://www.dbs.ifi.lmu.de/Publikationen/Papers/KDD-96.final.frame.pdf>

[1] DBSCAN



- ❖ DBSCAN algorithm learns a bidirectional (undirected) graph.
- ❖ Edges are formed differently by two parameters: EPS (ϵ) and minimum points.

[1] DBSCAN (closeness)

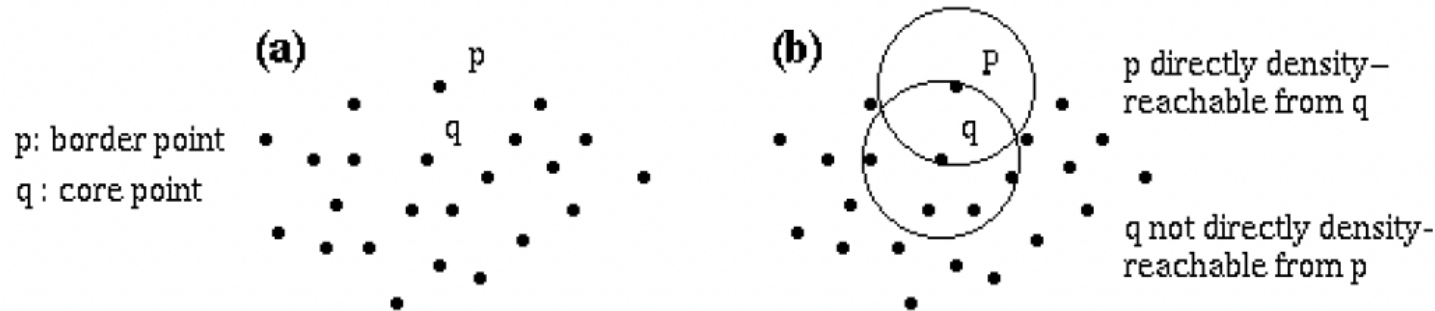
Definition 1: (Eps-neighborhood of a point) The *Eps-neighborhood* of a point p , denoted by $N_{\text{Eps}}(p)$, is defined by $N_{\text{Eps}}(p) = \{q \in D \mid \text{dist}(p,q) \leq \text{Eps}\}$.

[2] DBSCAN (closeness & sparsity: directly density reachable)

Definition 2: (directly density-reachable) A point p is *directly density-reachable* from a point q wrt. Eps , $MinPts$ if

- 1) $p \in N_{Eps}(q)$ and
- 2) $|N_{Eps}(q)| \geq MinPts$ (core point condition).

Obviously, directly density-reachable is symmetric for pairs of core points. In general, however, it is not symmetric if one core point and one border point are involved. Figure 2 shows the asymmetric case.

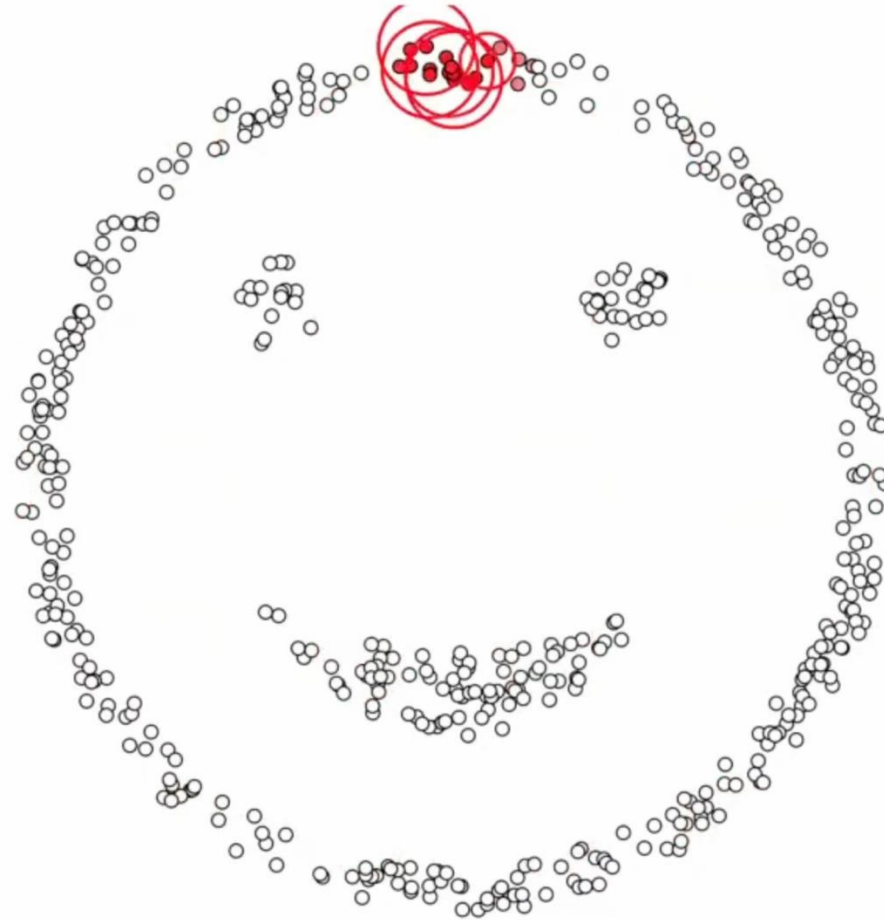


$q \rightarrow p$
: p is directly
researchable from q

[3] DBSCAN (algorithm)

1. Start with a core point “ p ” and form a new cluster.
2. All eps neighbors (p) are added to the cluster.
3. All eps neighbors (p) are marked as visited.
4. Expand the cluster by adding the neighbors of each of eps neighbors (p).
5. Repeat 3 and 4 until no more point to expand.
6. Explore another core points not visited before.
7. Any non-clustered points become noise.

[4] DBSCAN (example)

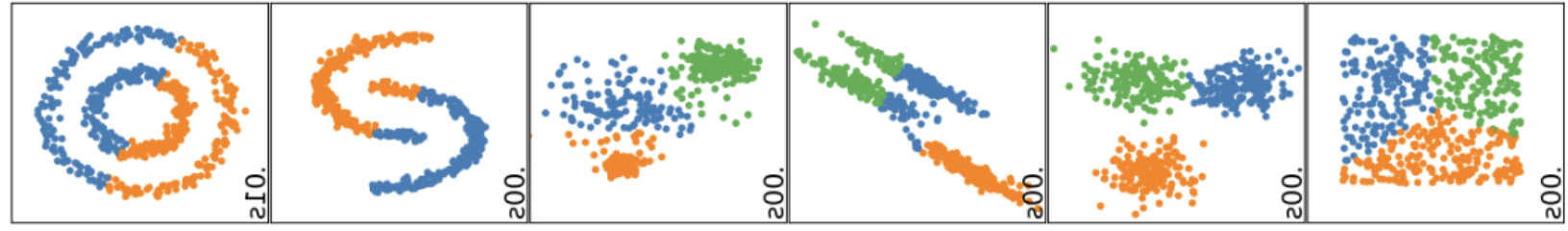


https://www.youtube.com/watch?v=_A9Tq6mGtLI&t=3s

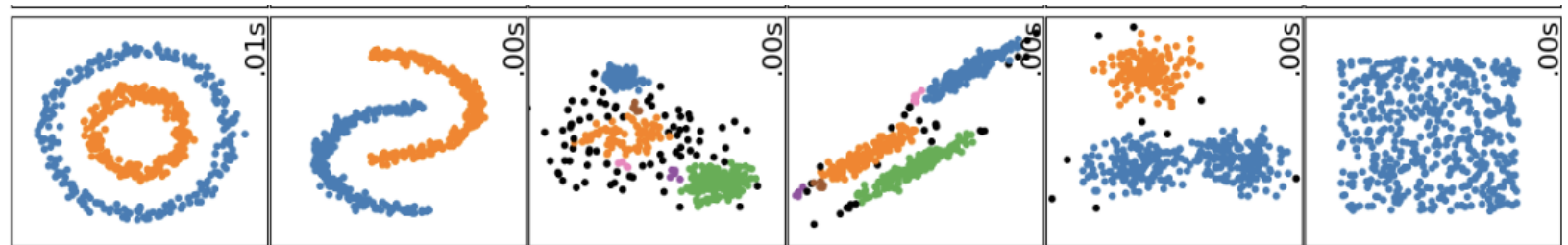
[5] DBSCAN (results K-means vs. DBSCAN, K=2)

https://scikit-learn.org/stable/auto_examples/cluster/plot_cluster_comparison.html#sphx-glr-auto-examples-cluster-plot-cluster-comparison-py

[K means]



[DBSCAN]



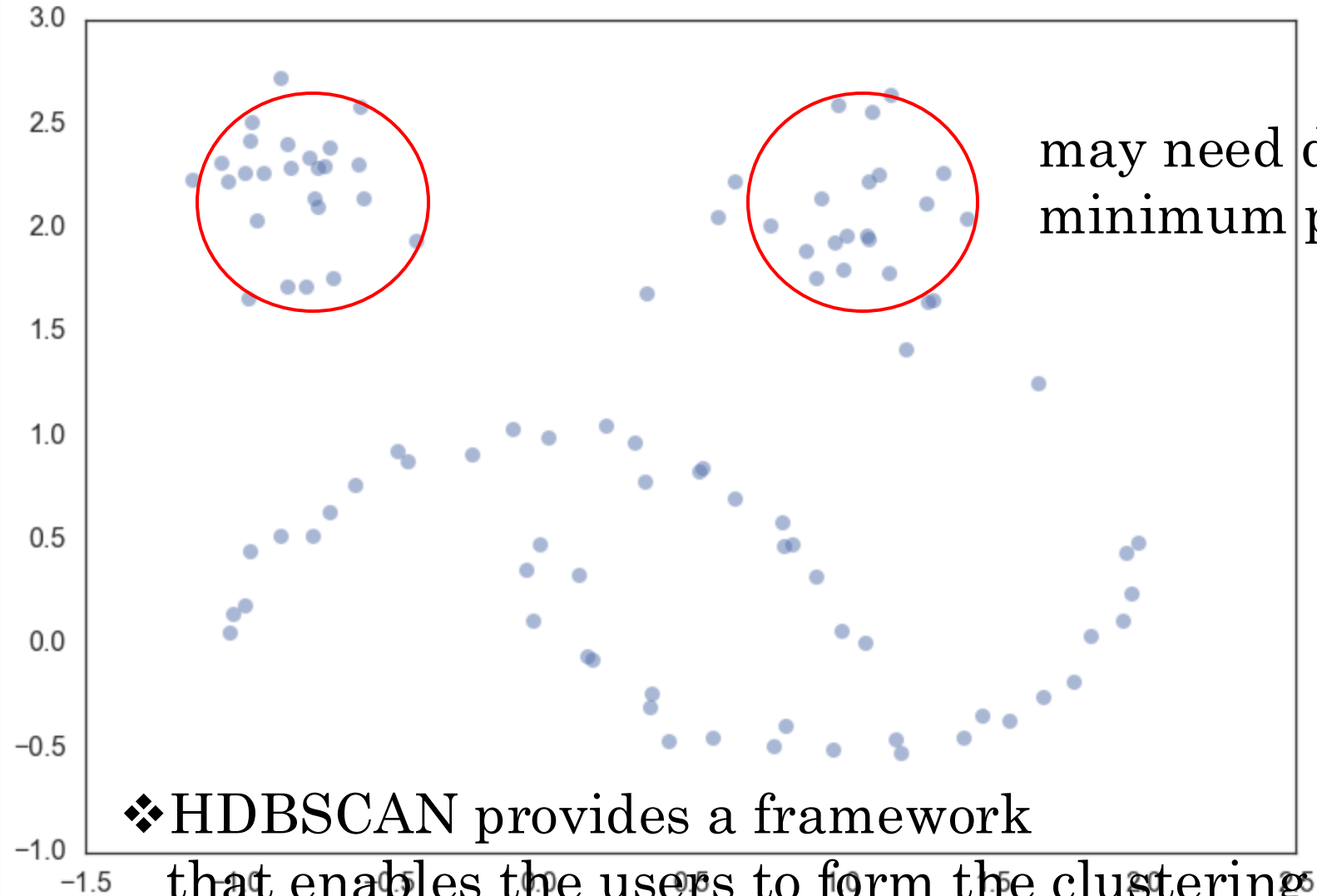
Q: can you point out the disadvantages of DBSCAN?

- Hierarchical DBSCAN

within data, some clusters have sparse connections while others have dense connections.
We need various water levels.

Library review: https://hdbscan.readthedocs.io/en/latest/how_hdbscan_works.html

[1] HDBSCAN



may need different eps and
minimum points condition.

❖ HDBSCAN provides a framework

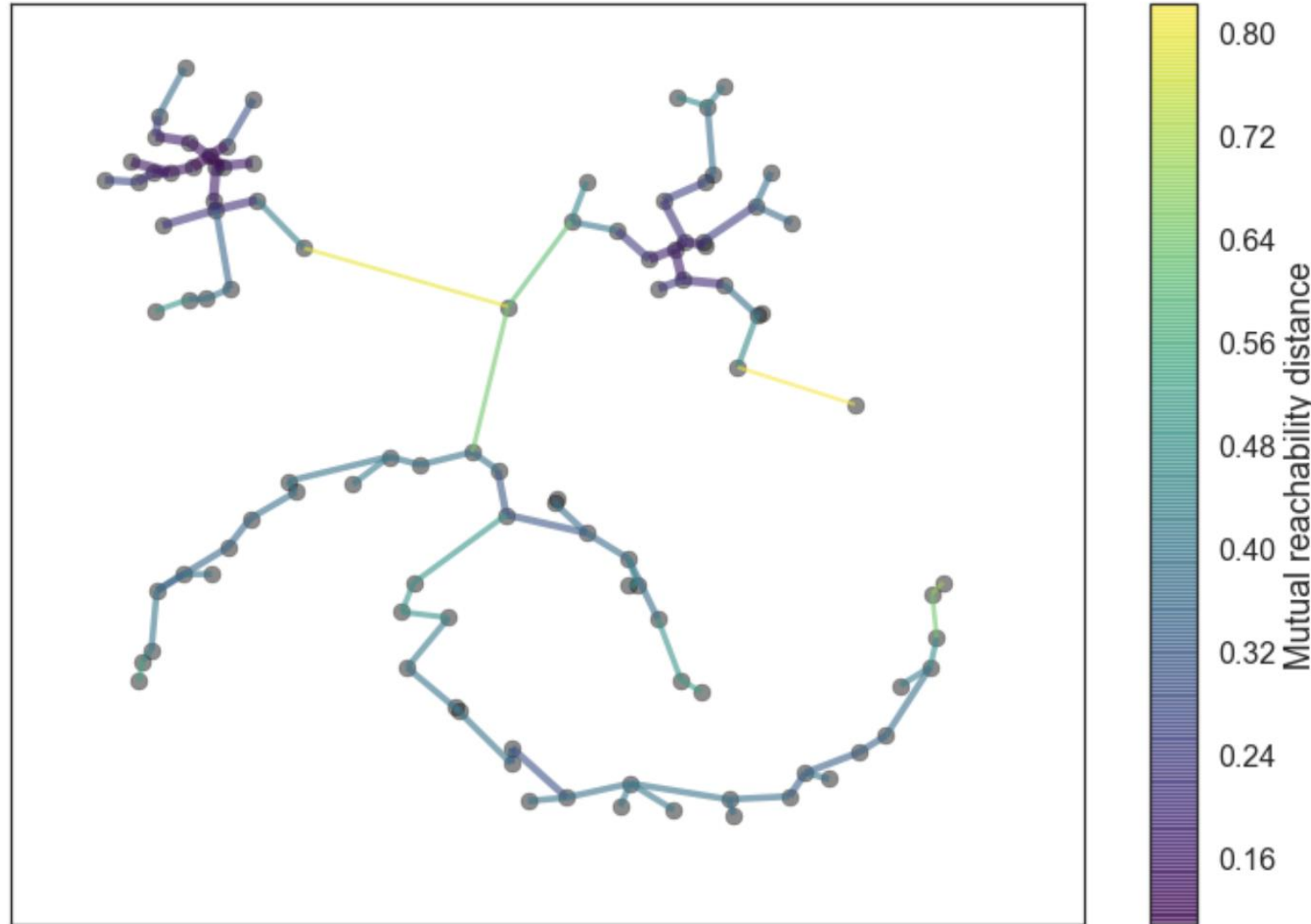
that enables the users to form the clustering graphs with various conditions for closeness and sparsity.

[2] HDBSCAN (closeness & sparsity)

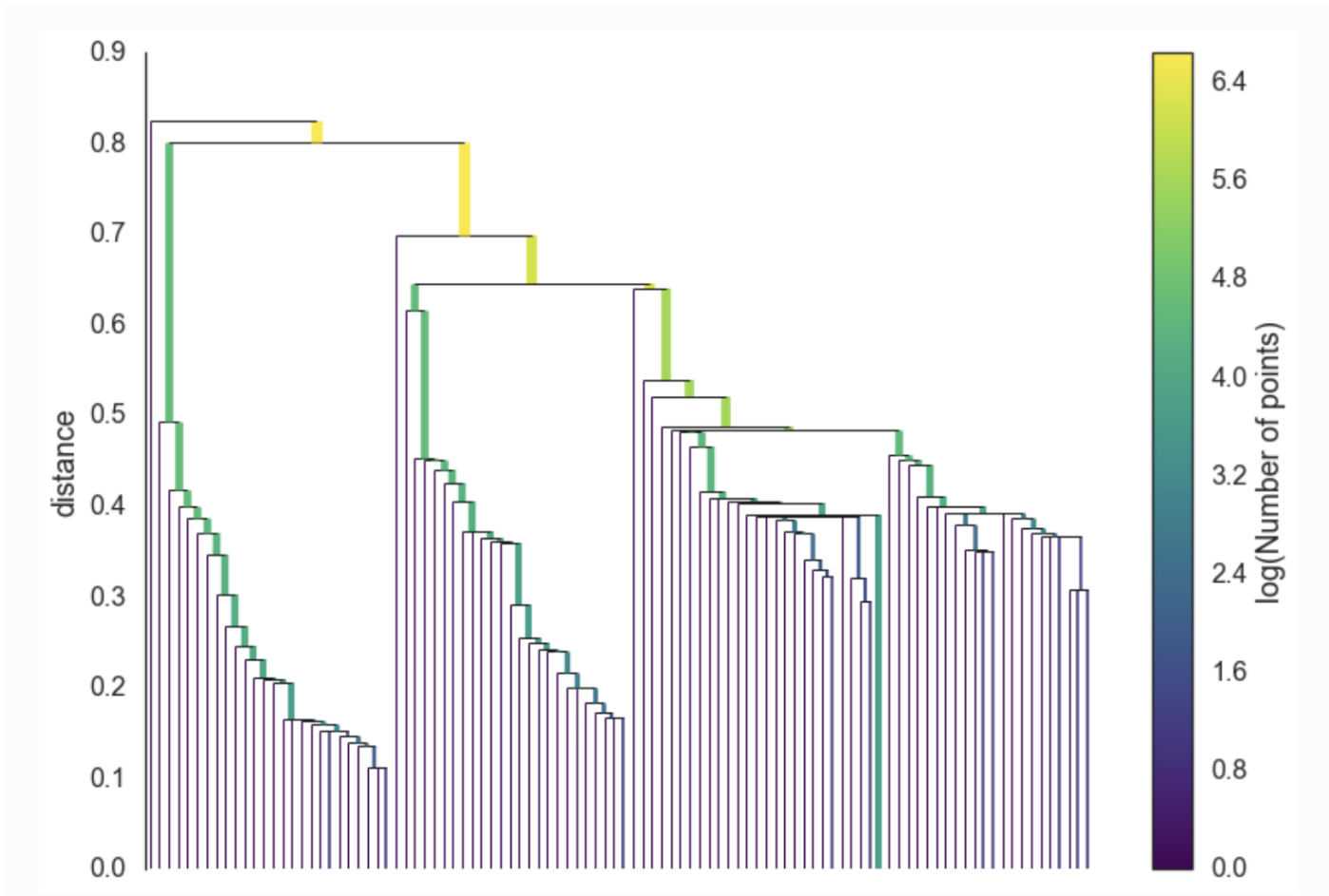
❖ Mutual Reachability Distance

$$d_{\text{mreach-}k}(a, b) = \max\{\text{core}_k(a), \text{core}_k(b), d(a, b)\}$$

[3] HDBSCAN (minimal spanning tree based on reachability distance)



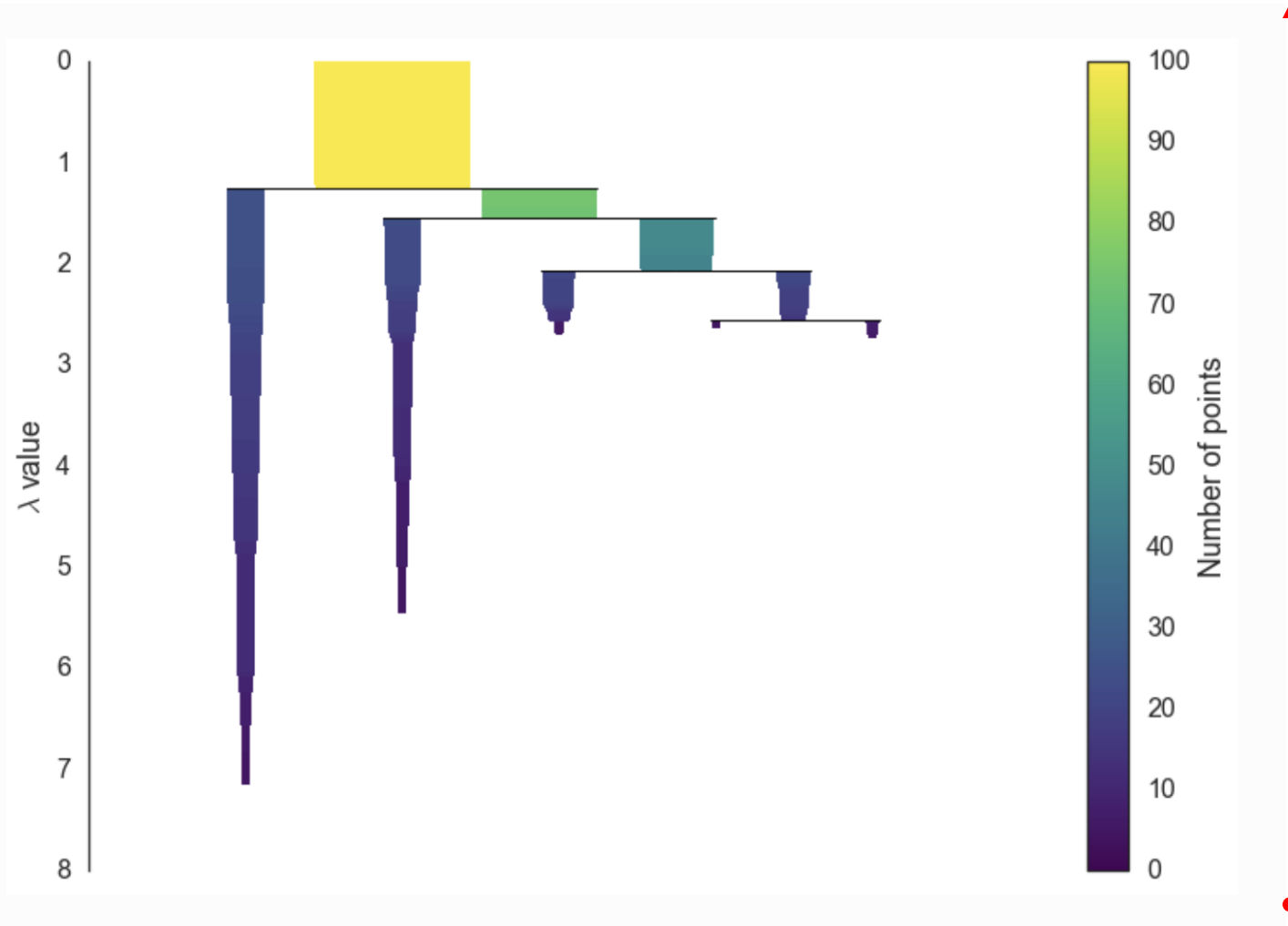
[4] HDBSCAN (building cluster tree, hierarchies)



■ water level is getting lower!

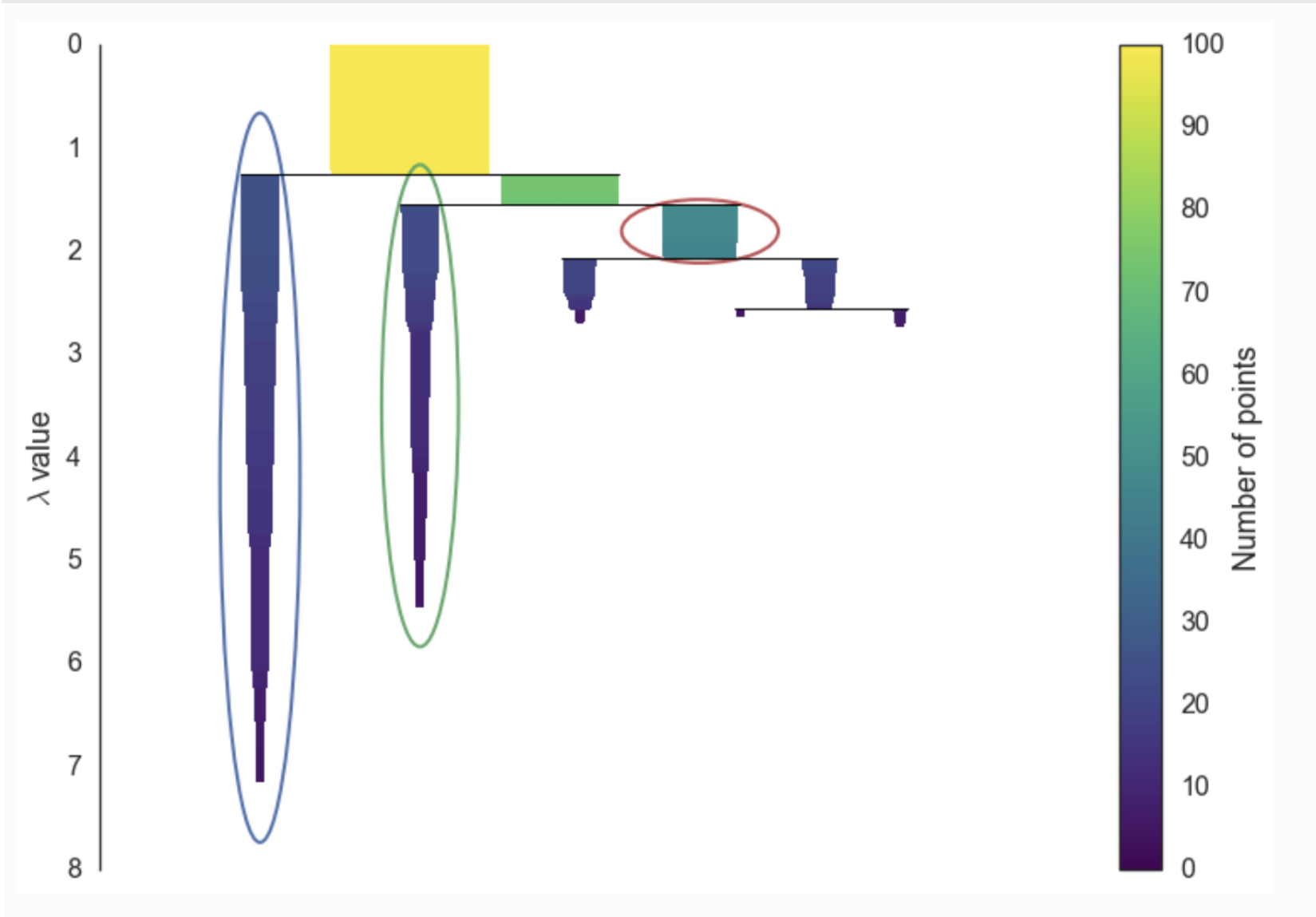
Q: Cutting at a specific distance gives us a set of clusters.
Do you think this is a good idea?

[5] HDBSCAN (condense the cluster tree)

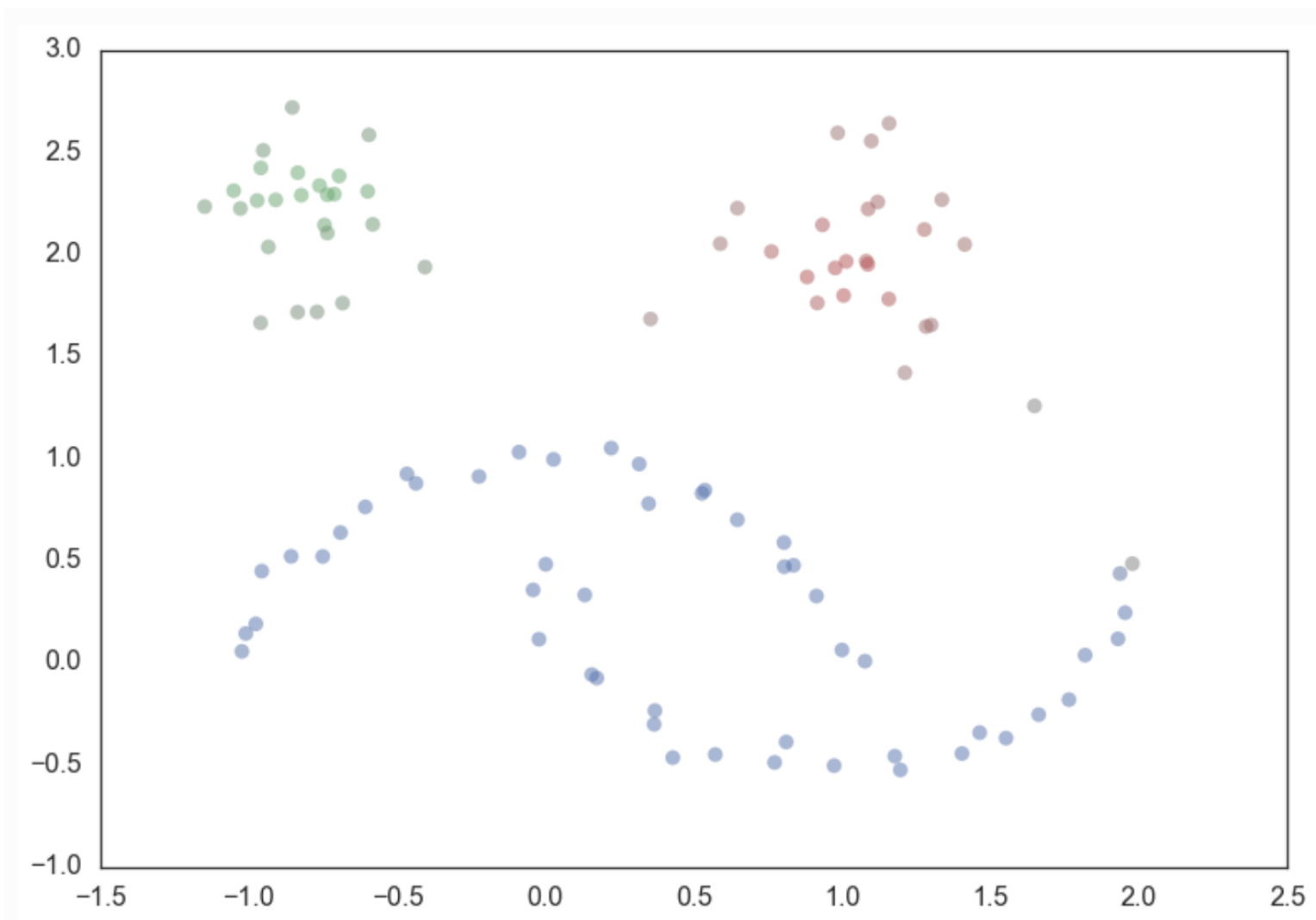


■ water level is getting lower!

[6] HDBSCAN (extract the clusters)



[7] HDBSCAN (clustering results)



Today's Lecture

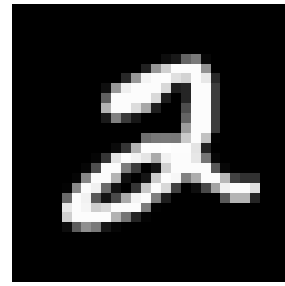
1. Representation Learning (various examples)
2. Probabilistic Modeling for Representation Learning
3. VAE Solution for the Intractability of Posterior
 - Loss Function: ELBO (Evidence Lower Bound)
 - Neural Net Implementation: Architecture, Loss, and Training Technique for end to end learning (reparameterization trick)
 - Experimental results with MNIST

- Data Representation

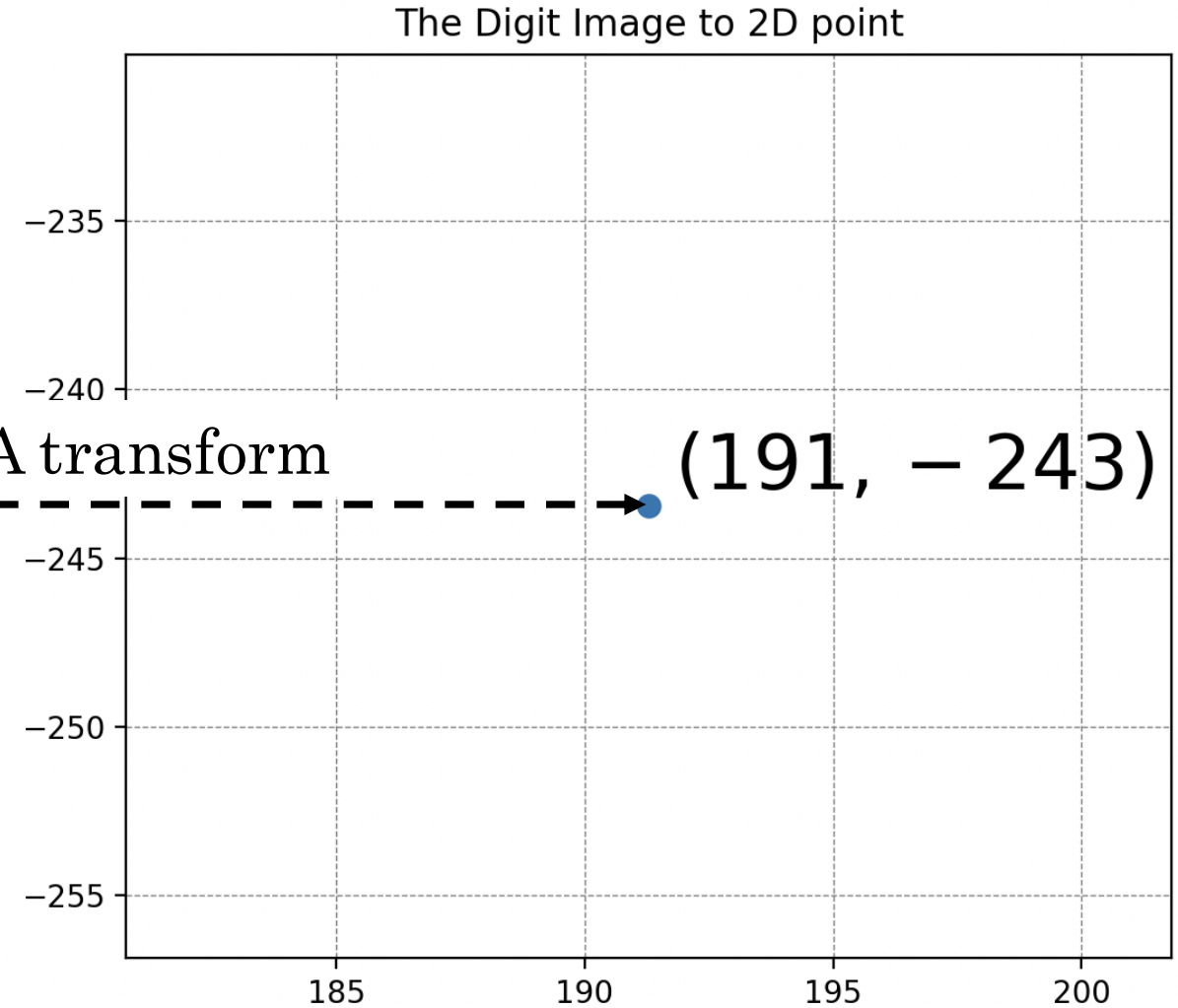
[1] Learning Data Representation

- Data representation is to transform **an original data sample** $\bullet \text{---} \rightarrow$ **to a new space.**

[1] Learning Data Representation (example1)



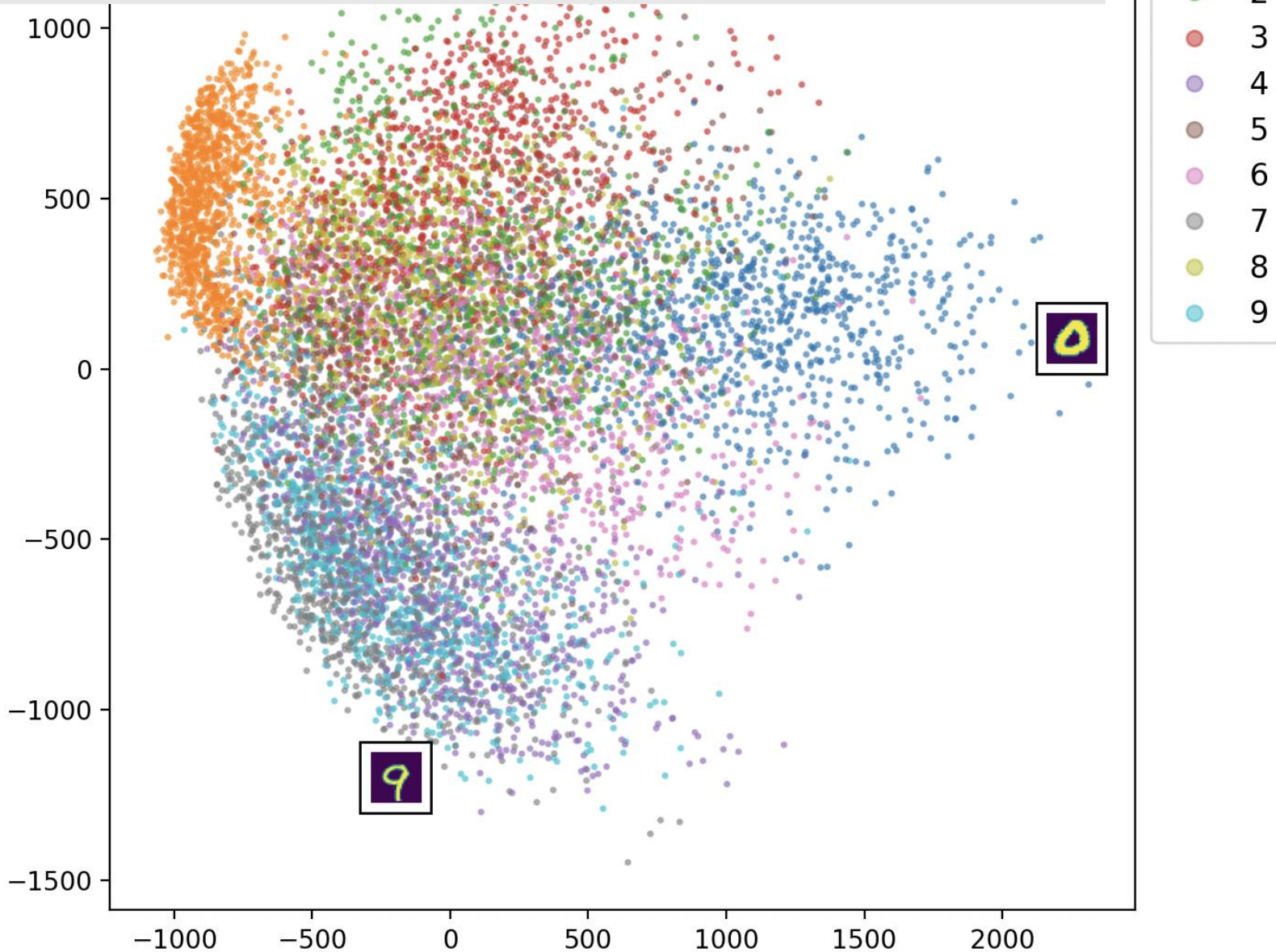
▪ by PCA transform



- the digit “2” ($28 \times 28 = 784$) is transformed to the 2D space by using PCA.

[2] Learning Data Representation (example1)

- the collective state of the PCA transformation

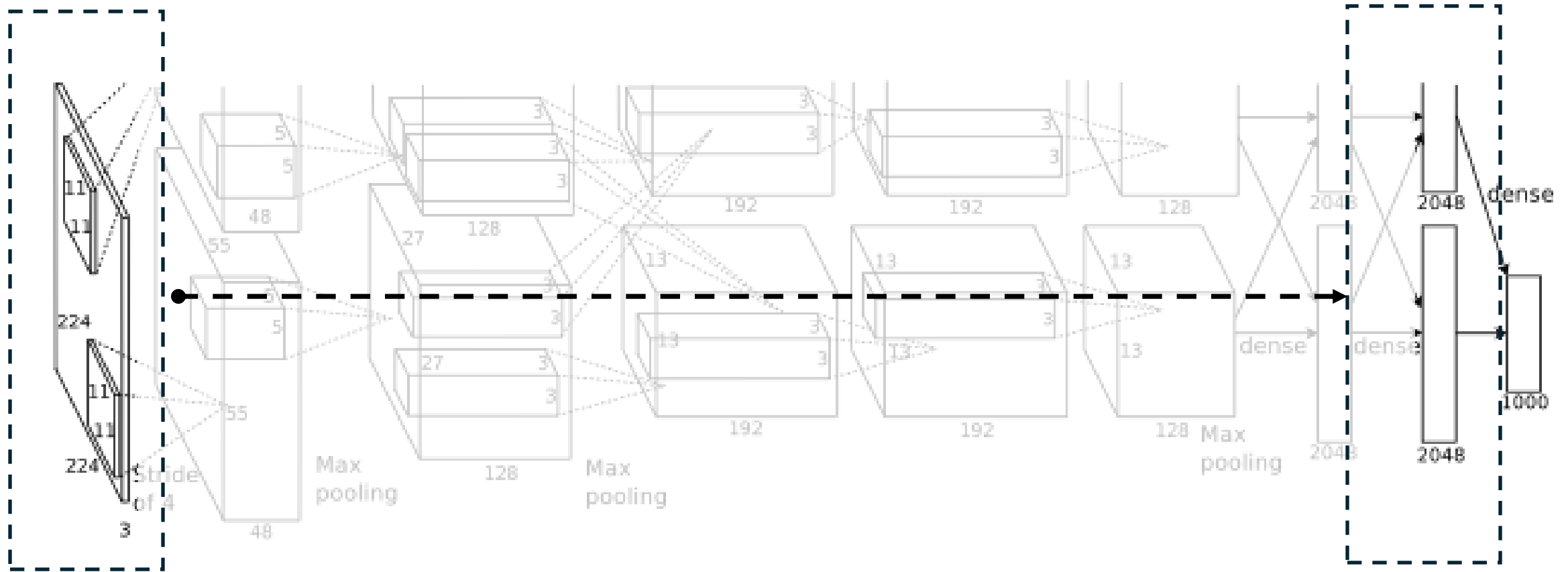


[3] Learning Data Representation

In machine learning,
it is important to learn a representation of data space.

- feature learning (classification)
- pattern discovery
- to generate new data samples (no exist before).

[4] Learning Data Representation (example1: feature learning)

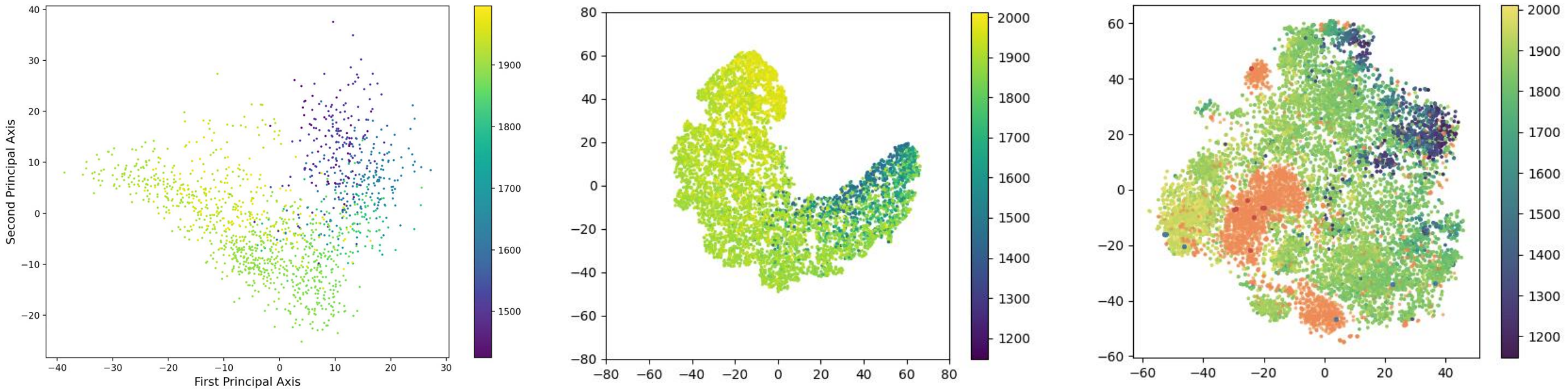


- Neural net learns a useful features for object classification.

From the original paper:

https://proceedings.neurips.cc/paper_files/paper/2012/file/c399862d3b9d6b76c8436e924a68c45b-Paper.pdf

[5] Learning Data Representation (example1: pattern discovery)



[Art Embeddings Color Coded by Year of Made]

- The latent space of various neural nets:
They reveal the collective / comparative / contrastive patterns of art samples.

[6] Learning Data Representation (example3: generative modeling)

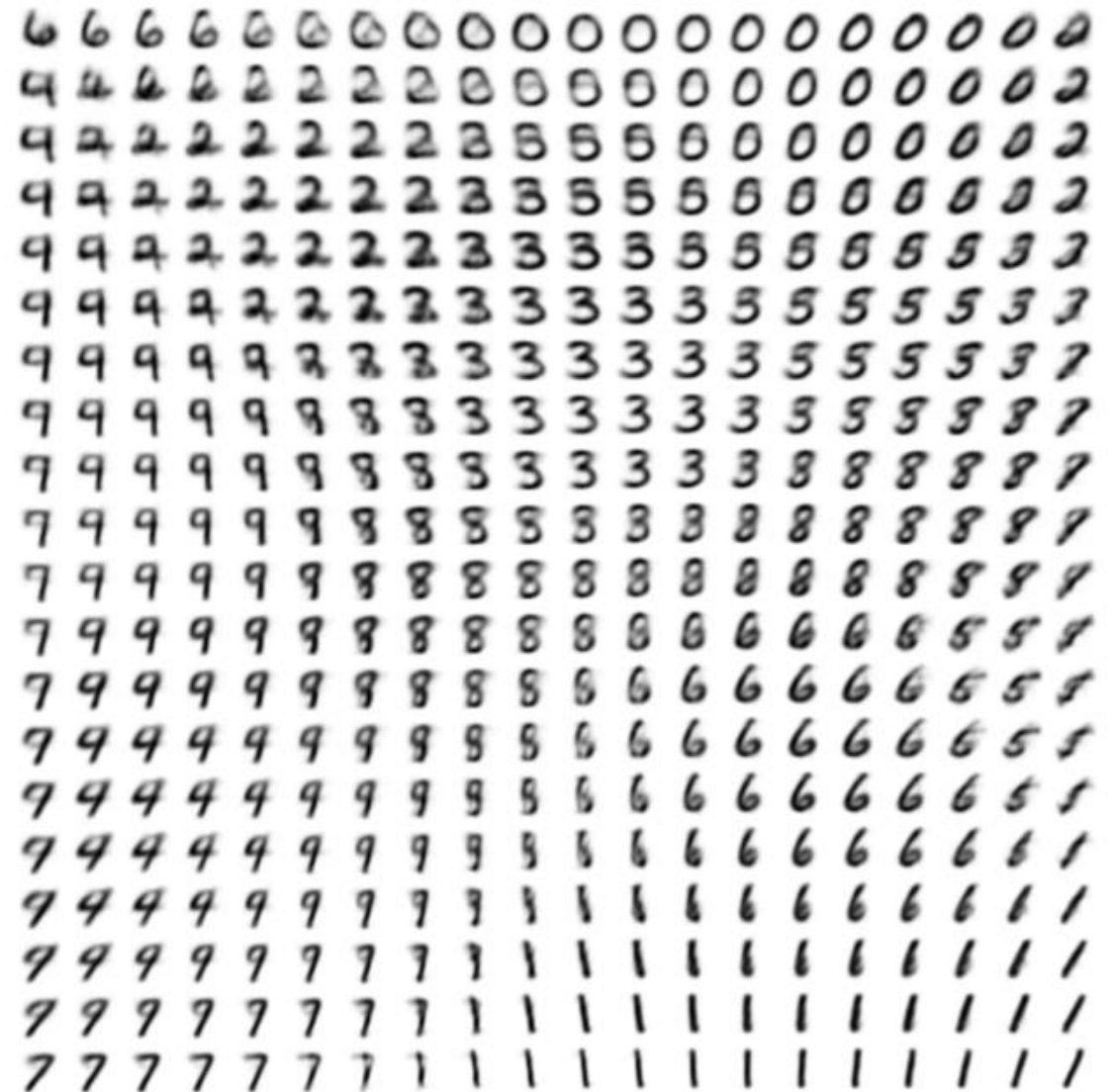


[Project Name: Generist Map]

- using interpolation of the representation space, we generate new data samples

[7] Learning Data Representation

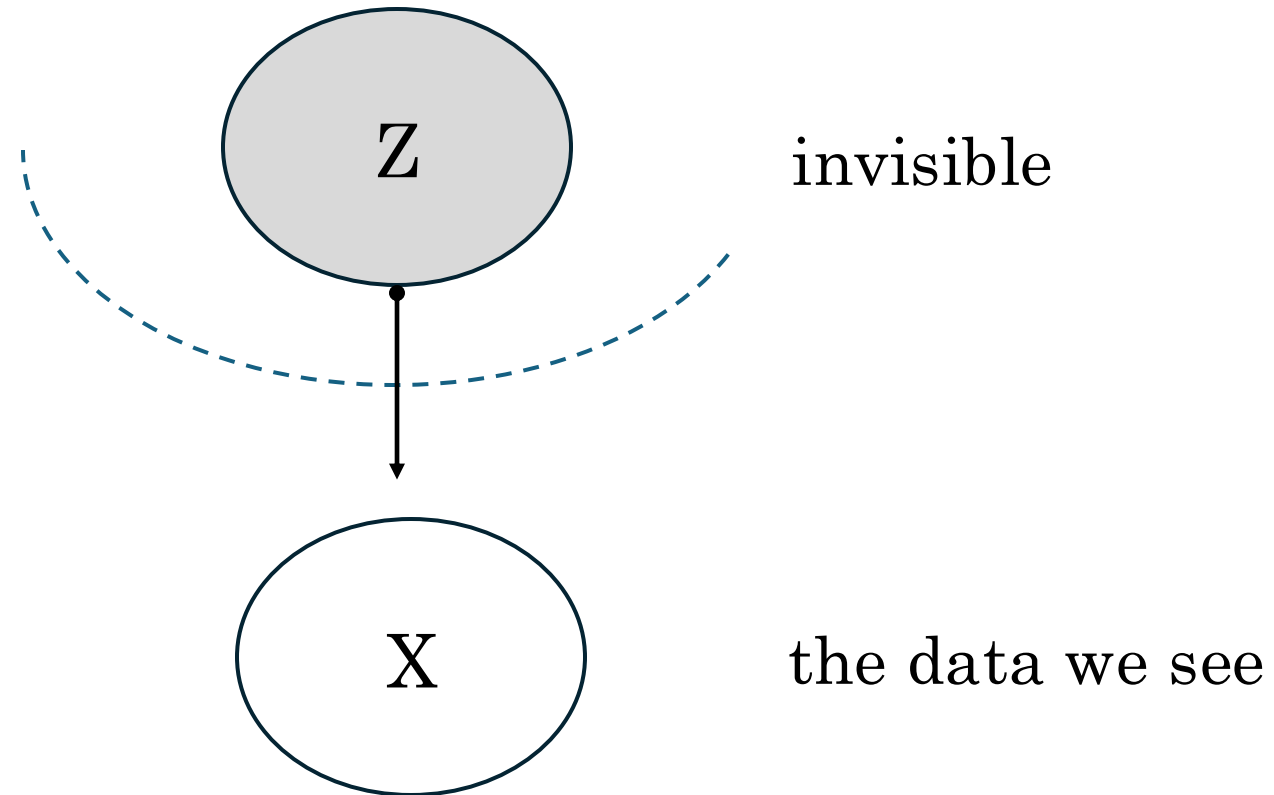
[2D VAE (Variational AutoEncoder)
[Representation of MNIST]



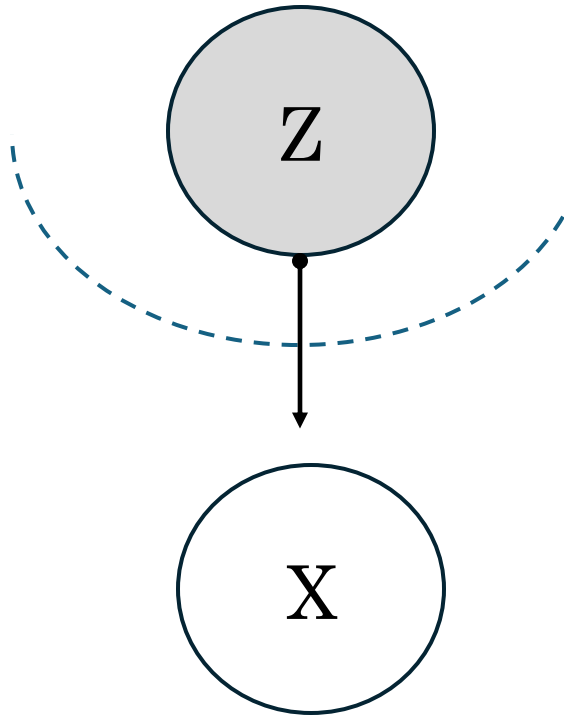
- Probabilistic Modeling for Representation

- Probabilistic Modeling for Representation
:Generative Modeling with Latent Variable

[1] Generative Modeling with Latent Variables

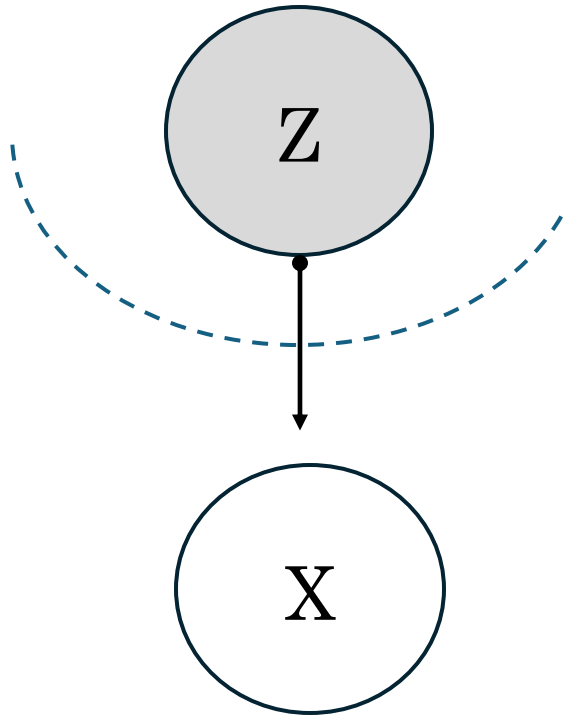


[2] Generative Modeling with Latent Variables



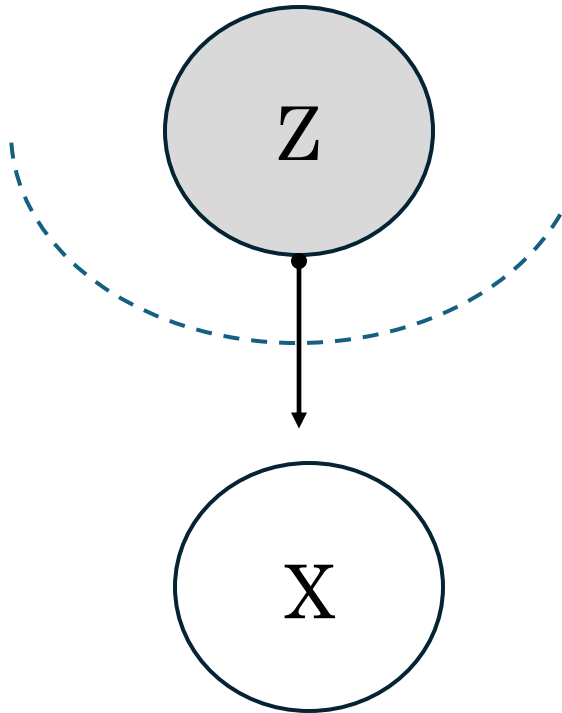
- $P(Z, X) = P(Z) \cdot P(X|Z)$
- Q: how this model is related to representation?

****terms...**



- $P(Z, X)$: joint density
- $P(Z, X) = P(Z) \cdot P(X|Z)$: prior density · conditional density
- $P(Z|X)$: posterior probability (reverse direction)
: representation learning

[3] Generative Modeling with Latent Variables



The problem of representation learning is translated into...

Q: how to learn $P(Z)$?
Q: how to learn $P(X|Z)$? } + joint density

[4] Generative Modeling with Latent Variables

“representation”: $P(Z|X)$

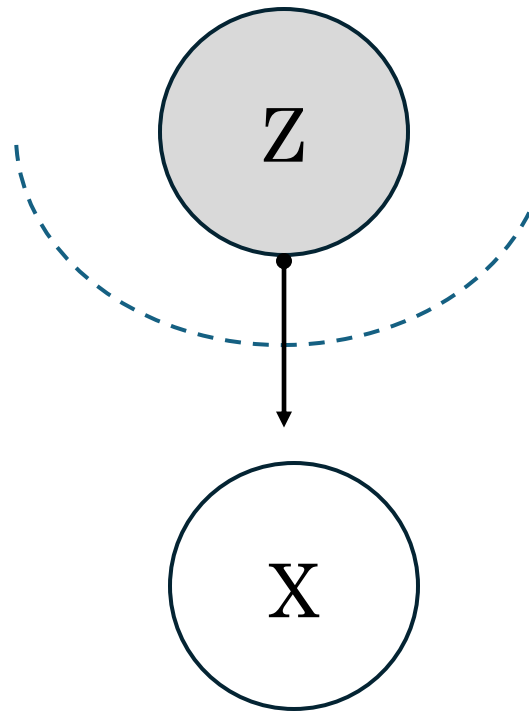


[5] Generative Modeling with Latent Variables

However, that may not be enough
Because often inference $P(Z|X)$ is intractable.

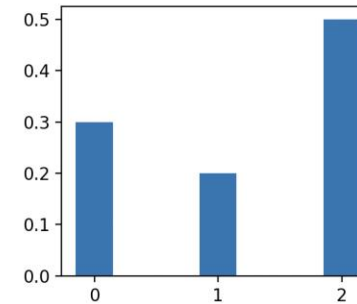


**Generative Modeling with Latent Variables



[Gaussian Mixture Model (GMM)]

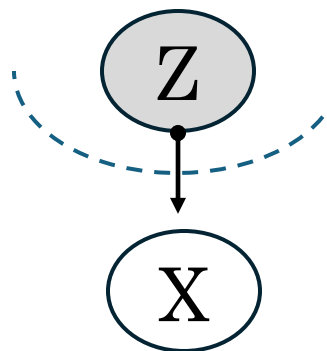
- $P(Z = k) = \pi_k$



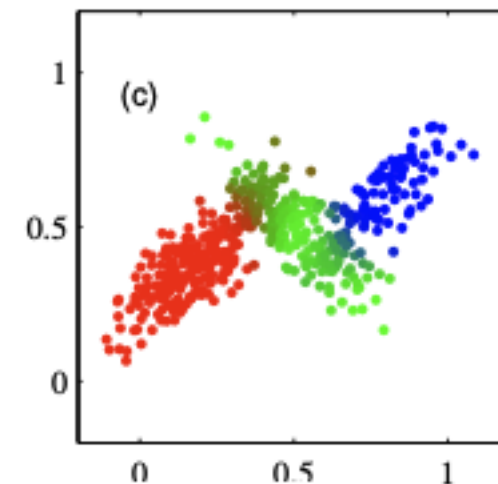
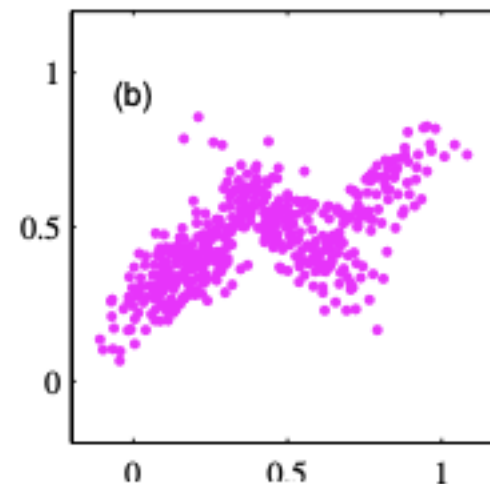
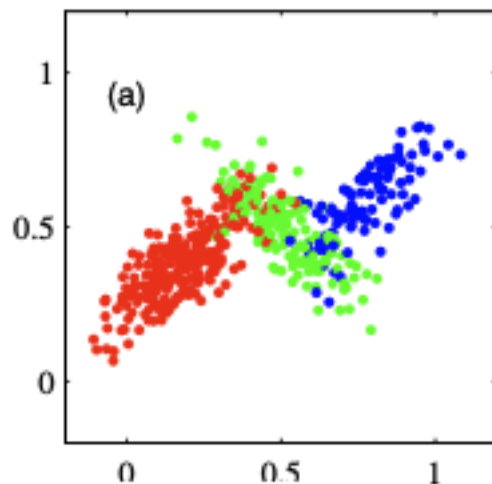
- $P(X|Z = k) = \mathcal{N}(\mu_k, \Sigma_k)$

from textbook Bishop Fig. 9.5

[6] Generative Modeling with Latent Variables



$$P(Z, X) = P(Z) \cdot P(X|Z)$$

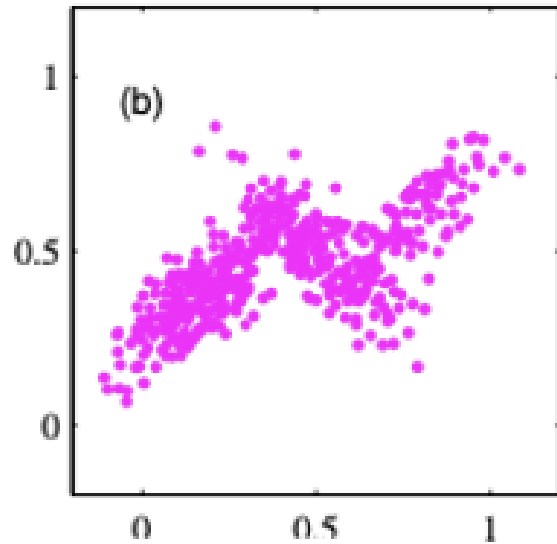


$P(Z|x)$

- representation will be $Z = \operatorname{argmax} P(Z|x)$
- the representation will be one of colors: red / green / blue.

from textbook Bishop Fig. 9.5

[7] Generative Modeling with Latent Variables



EM (**E**xpectation **M**aximization) to learn
 $P(Z)$ and $P(X|Z)$.

[8] Generative Modeling with Latent Variables

Then we can compute $P(Z|x)$,

It is easy to compute the posterior!

$$P(z|x) = \frac{P(z, x)}{P(x)} = \frac{P(x|z) \cdot P(z)}{\sum_z P(x|z) \cdot P(z)} = \frac{\pi_k \cdot \mathcal{N}(\mu_k, \Sigma_k)}{\sum_{Z=k} \pi_Z \cdot \mathcal{N}(\mu_Z, \Sigma_Z)}$$

from textbook Bishop Fig. 9.5

[9] Generative Modeling with Latent Variables

However,

It is often hard to define posterior density as a closed form.
when the hidden variables are continuous.

[10] Generative Modeling with Latent Variables

Posterior distribution $P[Z|X]$ is often intractable when Z is continuous.
(no closed / no analytical form)

- $f(z) \sim \mathcal{N}(0, 1)$

- $f(x|z) \sim \mathcal{N}(f(z), 1)$

- $f(z|x) = \frac{f(z)f(x|z)}{\int f(z)f(x|z)dz} = \frac{f(z)f(x|z)}{\int \frac{1}{\sqrt{2\pi}} \exp -\frac{z^2}{2} \frac{1}{\sqrt{2\pi}} \exp -\frac{(x - f(z))^2}{2} dz}$

+ no closed form!

[11] Generative Modeling with Latent Variables

the intractability is problematic,

- impossible to learn $P(Z)$ and $P(X|Z)$ using EM algorithm
- even assuming $P(Z)$ and $P(X|Z)$, to find the representation, still we need $P(Z|X)$.

[12] Generative Modeling with Latent Variables

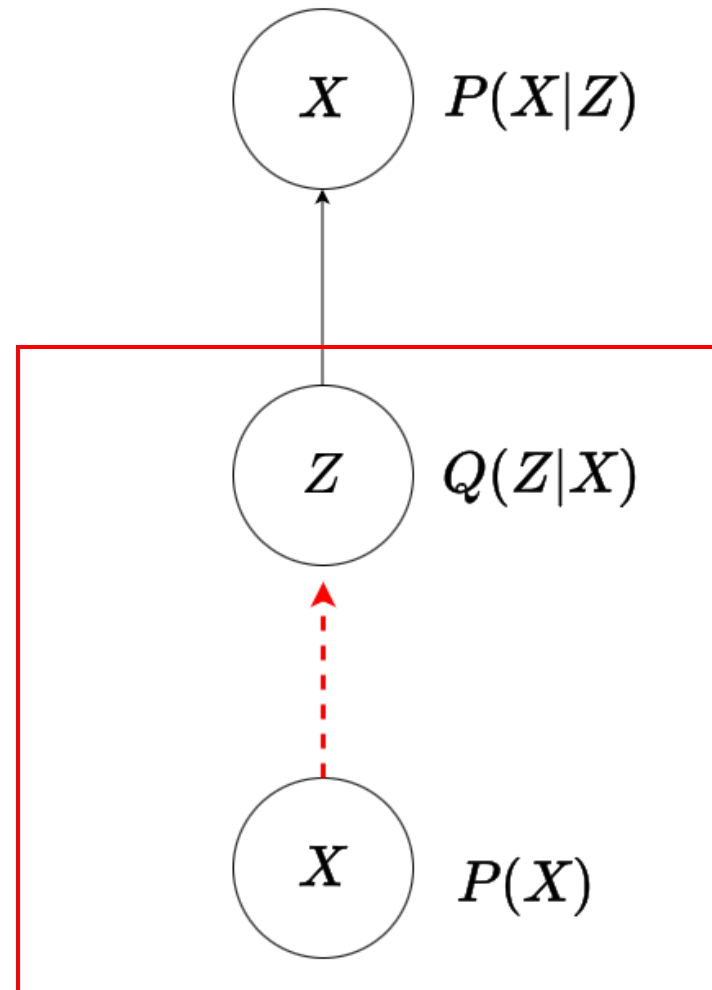
Q: How can we tackle the problem?

[13] Generative Modeling with Latent Variables

VAE finds the solution:

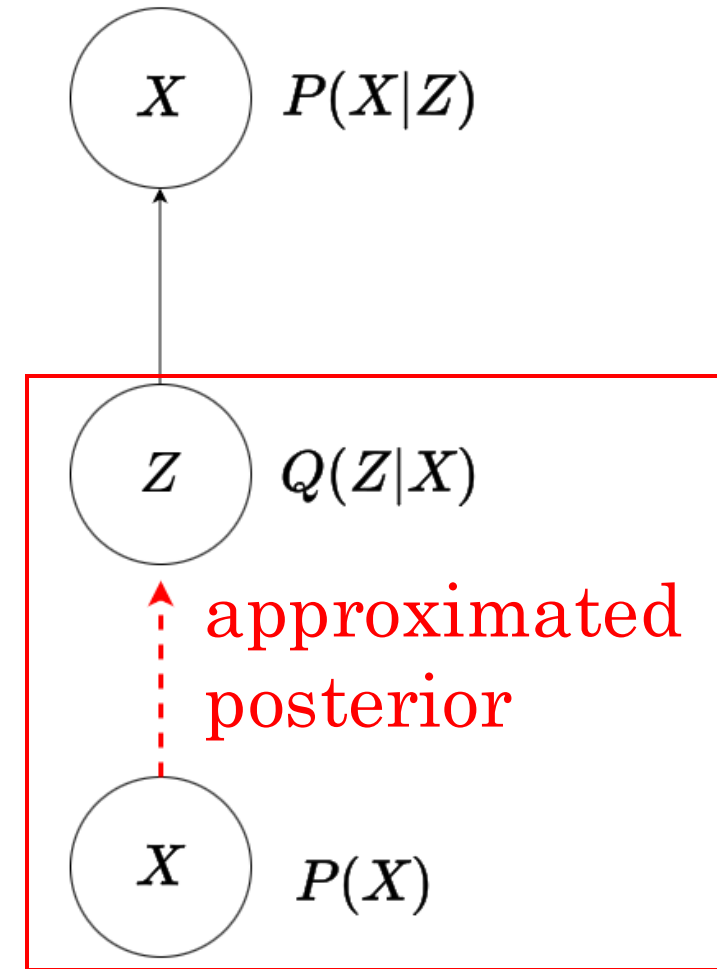
Q: what if we model $Q(Z|X)$ instead of $P(Z|X)$?
so we don't need to compute $P(Z|X)$ analytically.

[14] Generative Modeling with Latent Variables



[15] Generative Modeling with Latent Variables

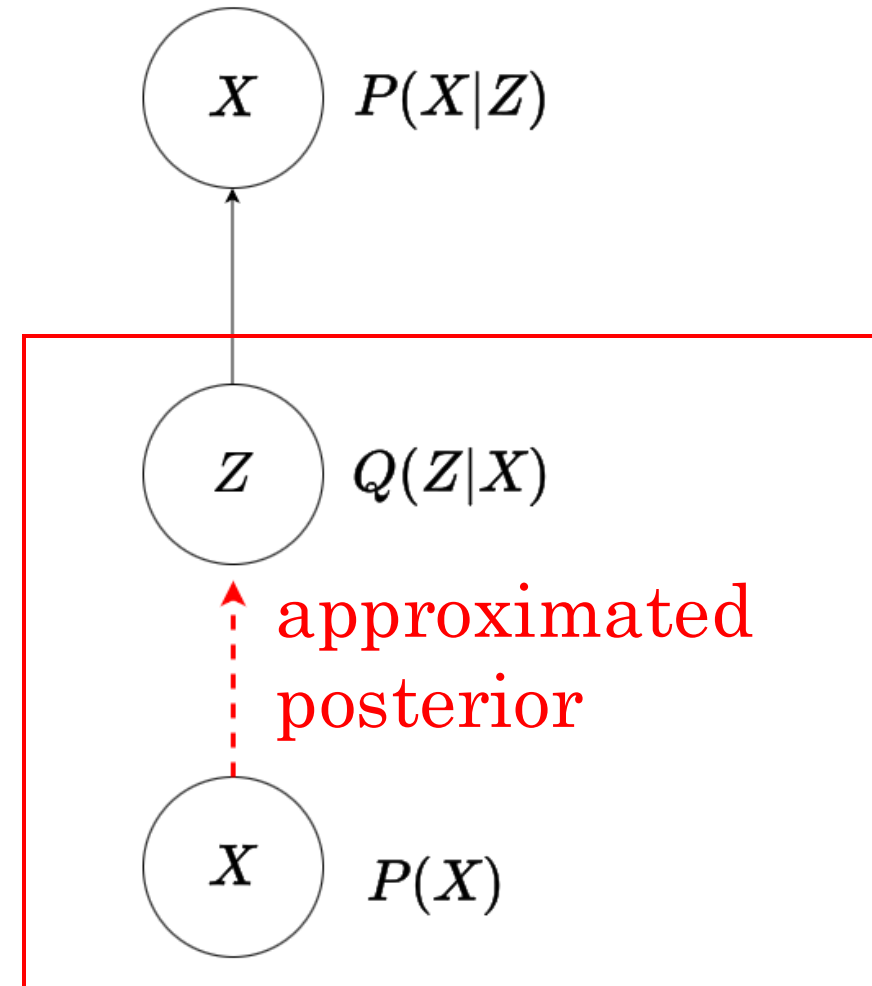
- $Q(Z|X)$ is **independent** from the original.
- $Q(Z|X) \neq P(Z|X)$ but we hope them get close.
(but don't need to be the same)



[16] Generative Modeling with Latent Variables

- we will learn the model with data.
the model needs to be **parameterized**.

- $P(X|Z, p)$
- $P(Z) \sim$ a prior
- $Q(Z|X, q)$



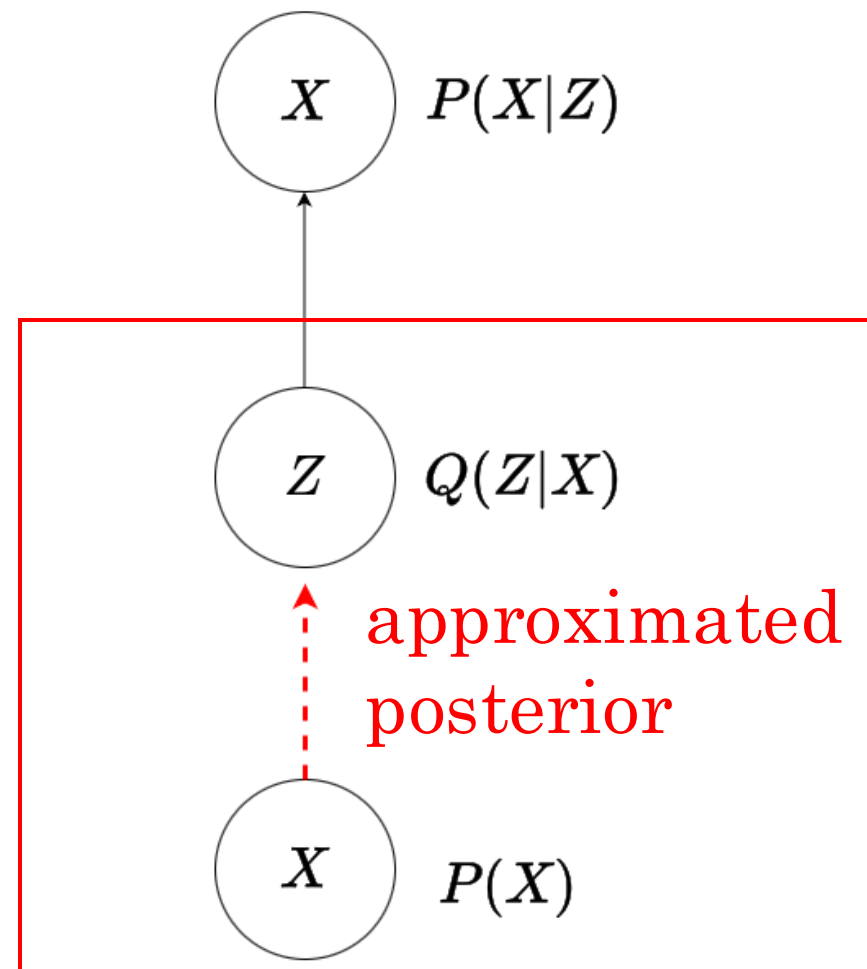
[17] Generative Modeling with Latent Variables

Q: how can learn the parameters?

we have the data and the model,
the components are parameterized.

Q: for training, what do we need?

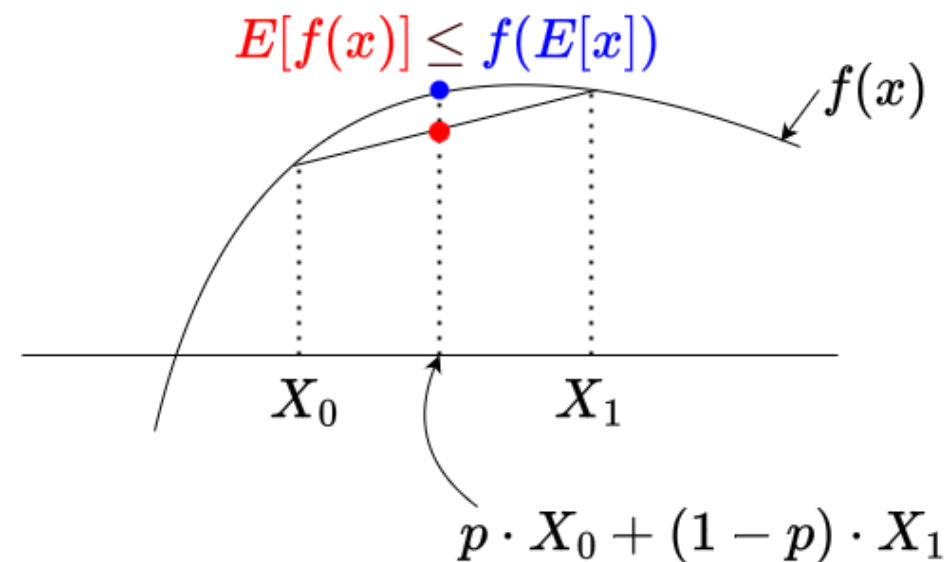
- $P(X|Z, p)$
- $P(Z) \sim$ a prior
- $Q(Z|X, q)$



- VAE Loss: **E**vidence **L**ower **B**ound (ELBO)

[1] ELBO

$$\begin{aligned}\log P(x|p) &= \log \sum_z P(z, x|p) \\ &= \log \sum_z \frac{P(z, x|p) \cdot Q(z|x, q)}{Q(z|x, q)} \\ &\geq E_{Q(z|x, q)} \left[\log \frac{P(z, x|p)}{Q(z|x, q)} \right] \leftarrow \text{ELBO}\end{aligned}$$



[Jensen's Inequality]

[2] ELBO

[**E**vidence **L**ower **B**ound (ELBO)] $E_{Q(z|x,q)} \left[\frac{P(x, z|p)}{Q(z|x, q)} \right]$

the function is the lower bound of the log likelihood: $\log P(x|p)$.

****gap: $\log P(x|p) - \text{ELBO}$**

- $\log P(x|p) - \text{ELBO} = D_{KL}(Q(z|x, q) || P(z|x, p))$

$$\log P(x|p) - \sum_z \log \frac{P(x, z|p)Q(z|x, q)}{Q(z|x, q)}$$

$$= \sum_z Q(z|x, q) (\log P(x|p) - \log \frac{P(x, z|p)}{Q(z|x, q)})$$

$$= \sum_z Q(z|x, q) \log \frac{Q(z|x, q)}{P(z|x, p)}$$

$$= D_{KL}(Q(z|x, q) || P(z|x, p))$$

[3] ELBO

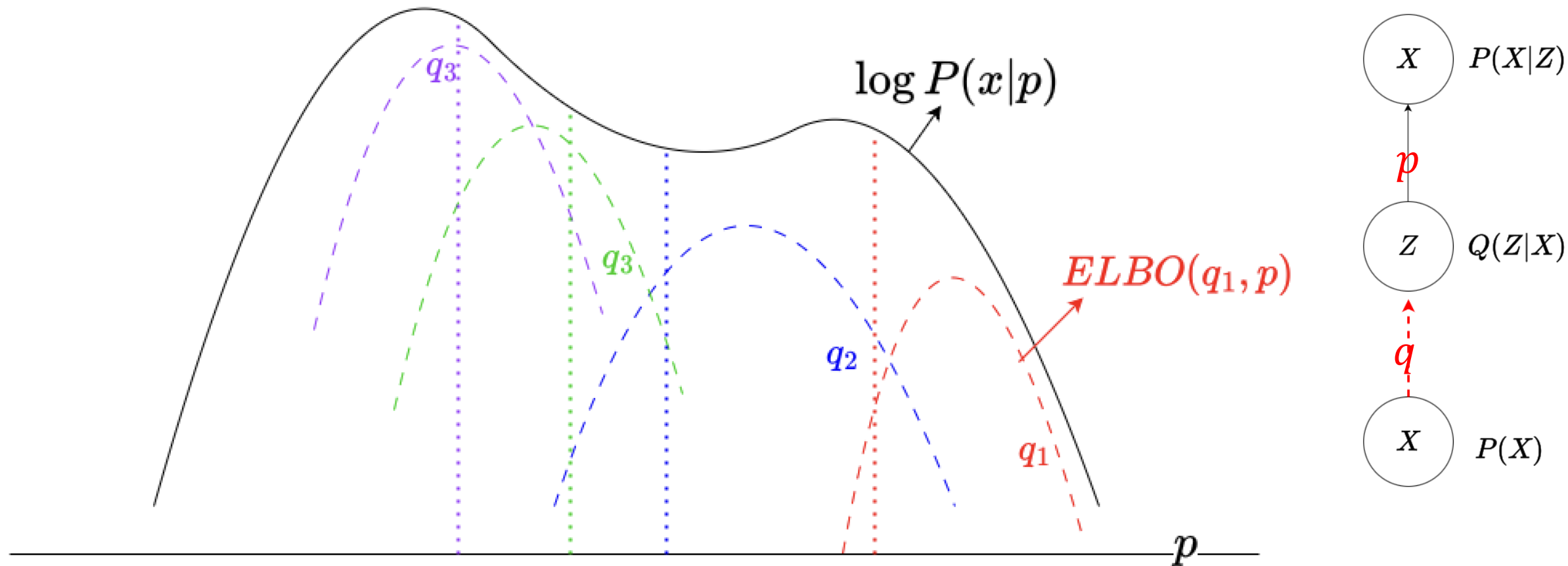
[some important points about ELBO]:

- $\log P(x|p)$ is the **function of p (the original) when data x given**
- $E_{Q(z|x,q)}[\frac{P(x,z|p)}{Q(z|x,q)}]$ is **the function of p and q (the original + approximation)**
- the gap between $\log P(x|p) = D_{KL} [Q(z|x,q)||P(z|x,p)] + E_{Q(z|x,q)}[\frac{P(x,z|p)}{Q(z|x,q)}]$

[4] ELBO

- VAE training is to increase ELBO $E_{Q(z|x,q)}[\frac{P(x,z|p)}{Q(z|x,q)}]$ by updating p and q .

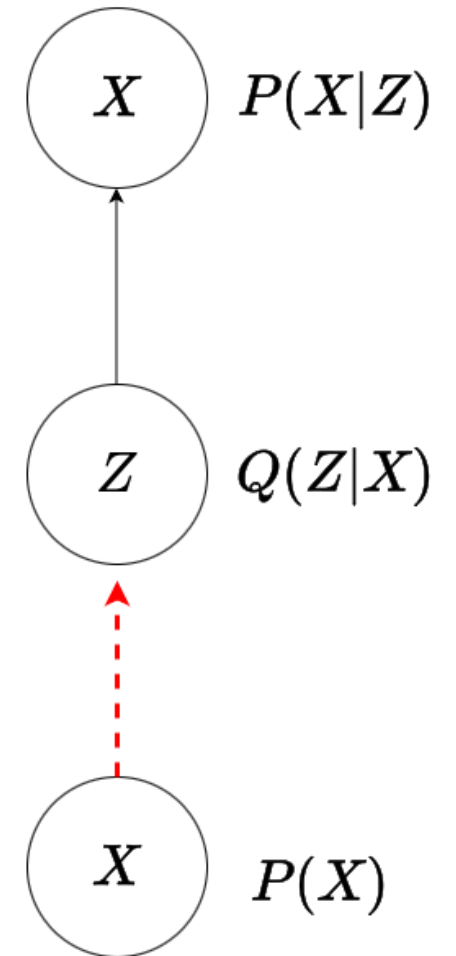
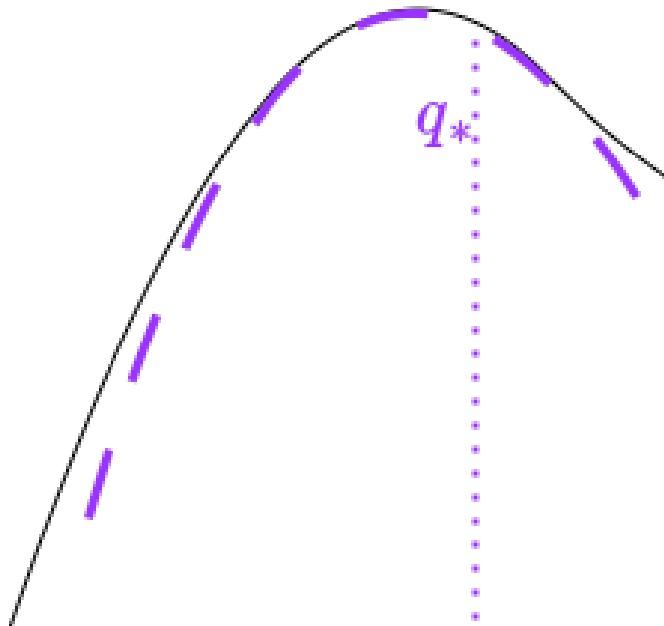
[5] ELBO (illustrative example)



- VAE training with ELBO
is to iteratively find a path (p) to increase $\log P(x|p)$.

[7] ELBO

Q: what if $Q(Z|X, q)$ have a high expressive power.
so no gap between $Q(Z|X, q^*)$ & $P(X|Z, p^*)$?



[8] ELBO

Hmm... that sounds great,
but is this really good to have the perfect fit to $P(Z|X, p)$?

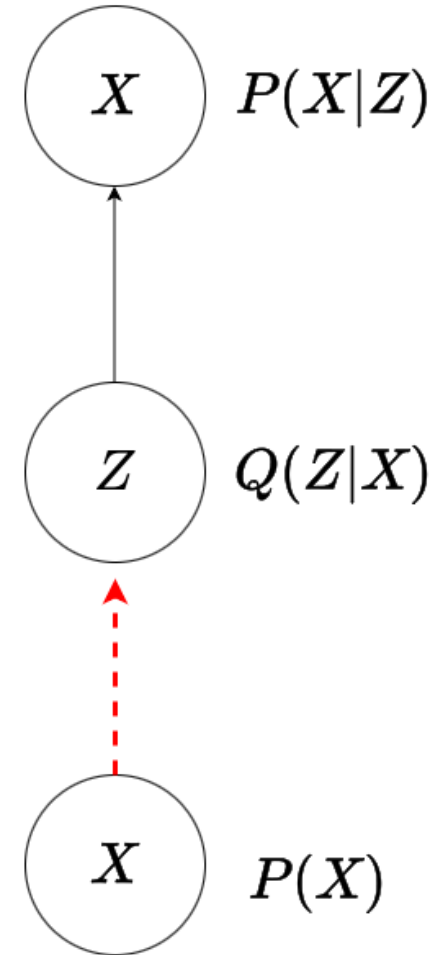
[9] ELBO



- how can design the high expressive $Q(Z|X)$?
- **how can we express ELBO in an algebraic formula.
- we want to extract salient features in representation.
(compact and structured)

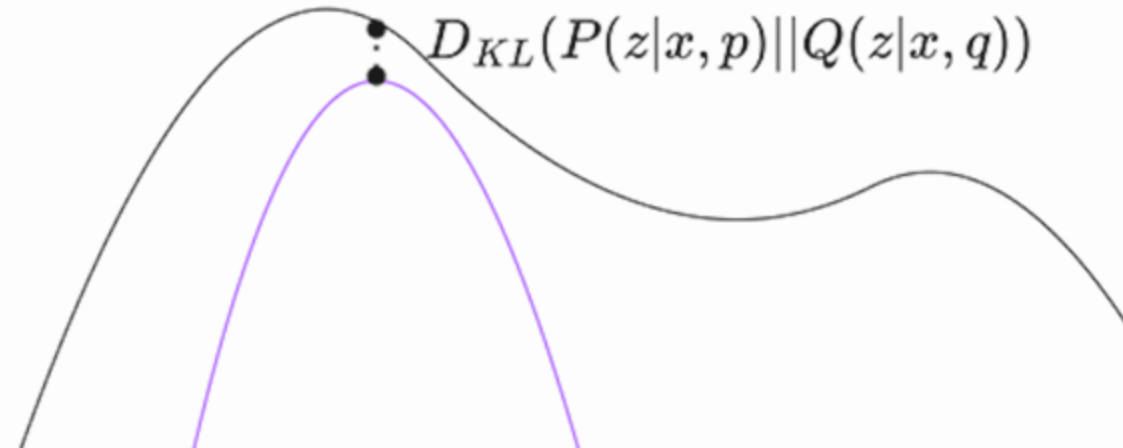
[10] ELBO

- by that reasons,
VAE constrains the model $Q(Z|X, q)$
with a tractable modeling (family).



[11] ELBO

[VAE training]



within the constraint that $Q(Z|X, q)$ has ,

- it aim to approximate $\log P(Z|X, p^*)$
- it reduces the gap $Q(Z|X, q)$ & $P(Z|X, p^*)$ as much as possible.

This technique is called **Variational Inference & Optimization**.

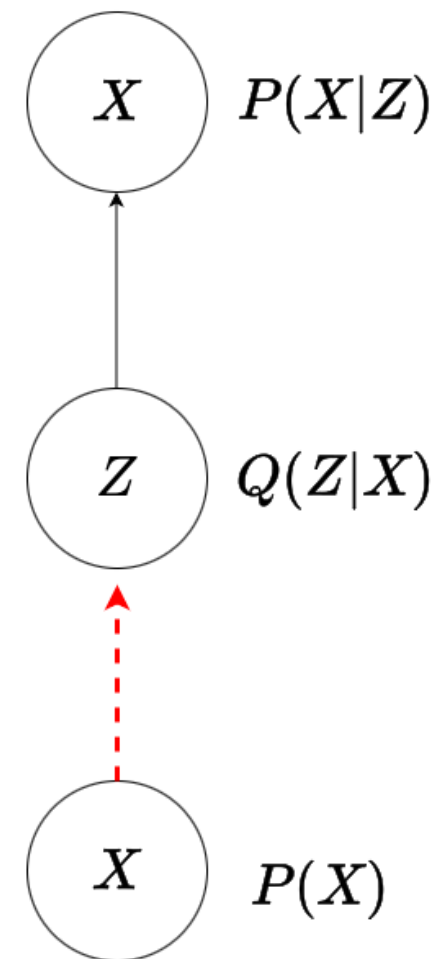
- by maximizing the lower bound: *the function of p and q*
- we aim to approximate $P(x|p^*)$ “ $P(z|x, p^*) \approx Q(z|x, q)$ ”

[13] ELBO

Q: how can learn the parameters?
we have the data and model, what do we need?

- $P(X|Z, p)$
- $P(Z) \sim$ a prior
- $Q(Z|X, q)$

A: by maximizing ELBO!

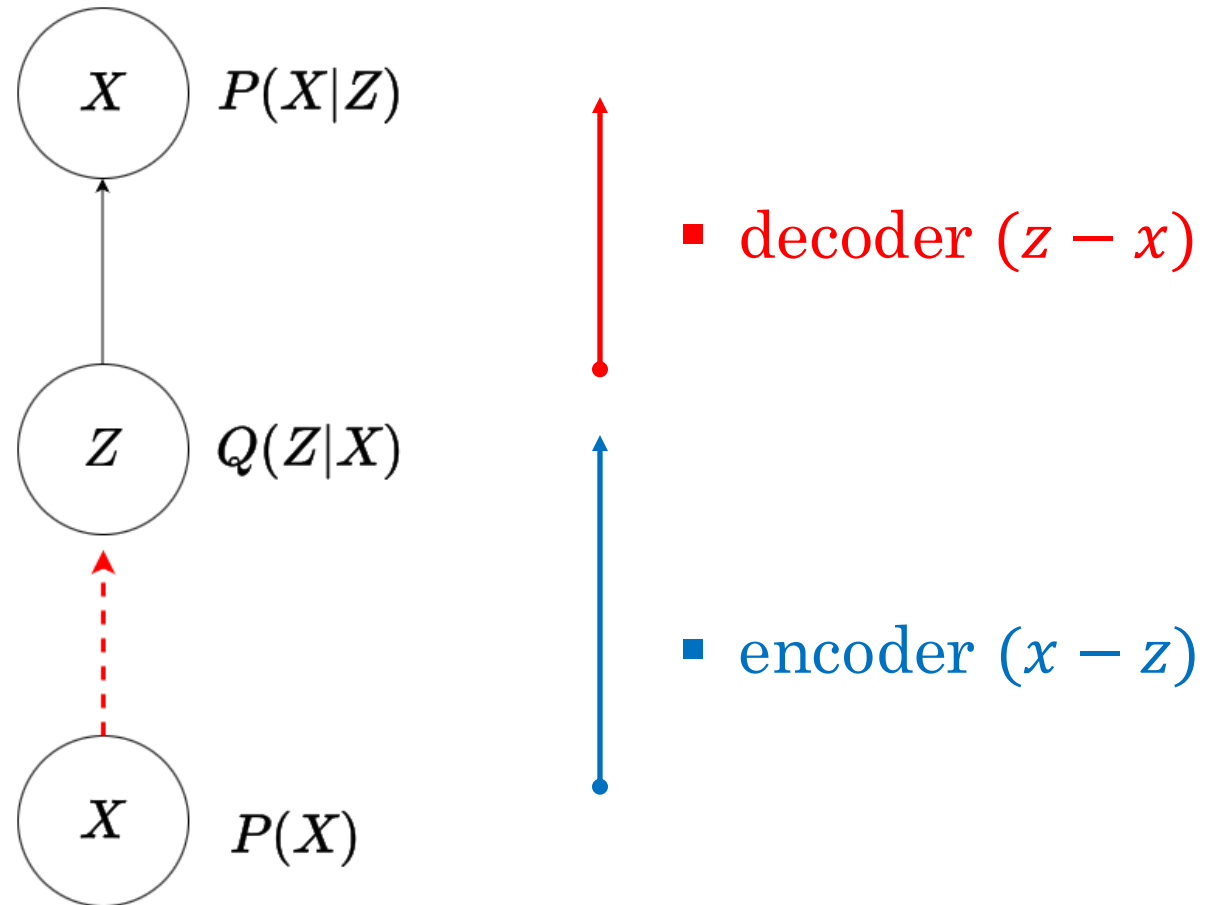


- VAE (Variational Auto-Encoder): abstract version

- (1) new addition $Q(Z|X, q)$
- (2) $Q(Z|X, q)$ is a tractable family.
- (3) ELBO Loss (p, q)

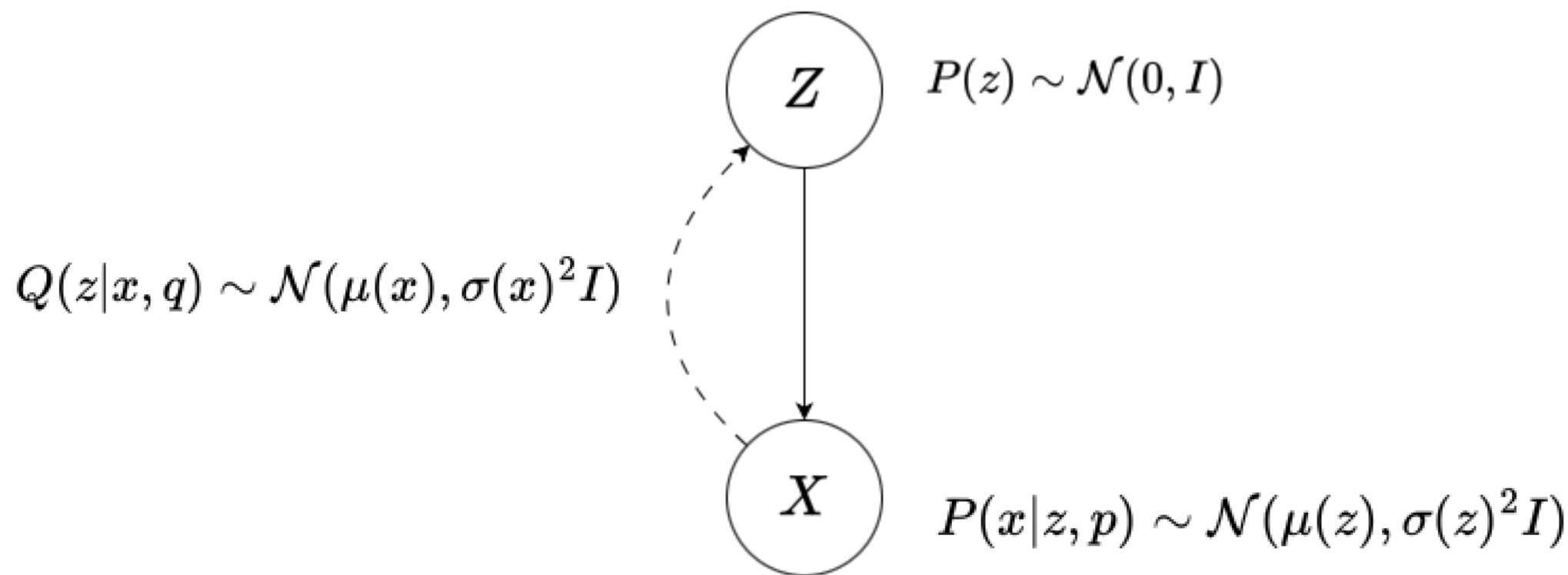
[1] VAE

- VAE structure: “**representation + generative**” in a unified framework!

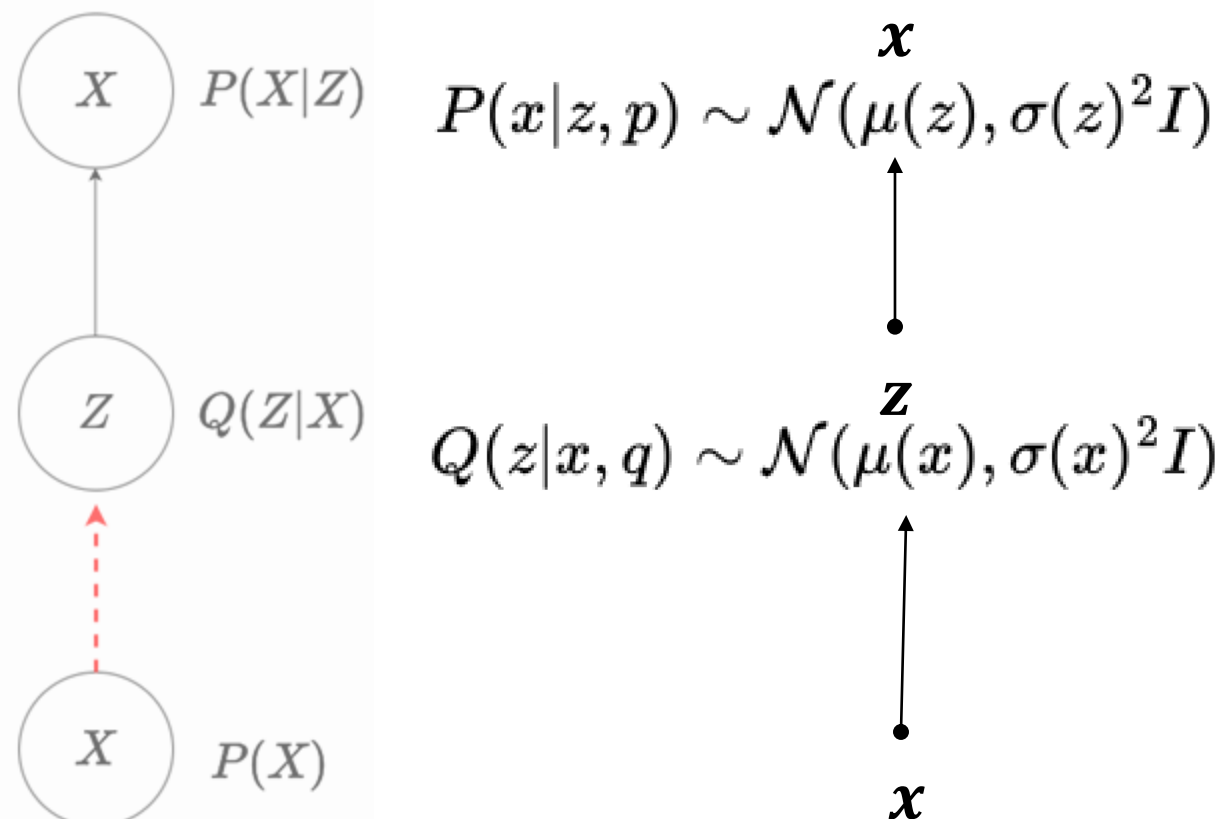


[2] VAE

- VAE defines the tractable densities.

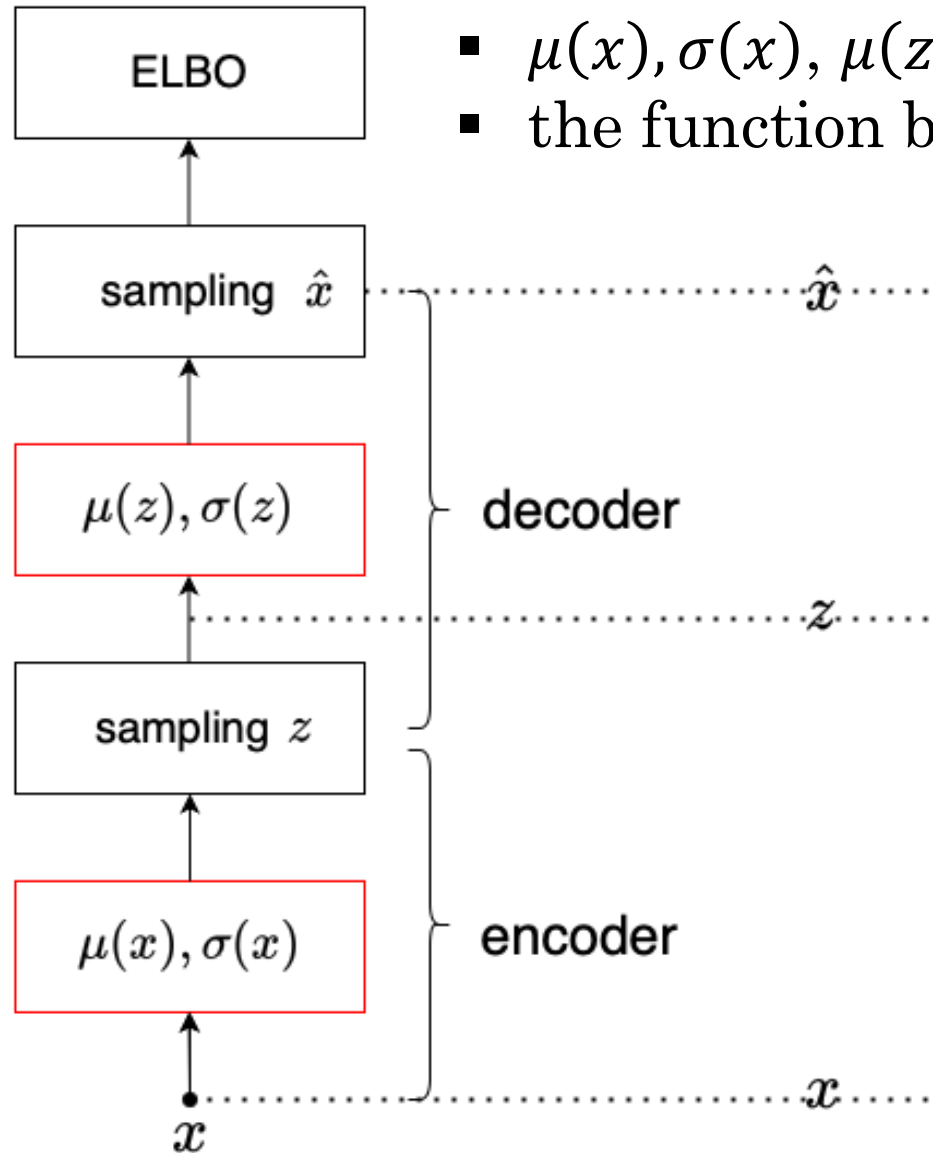


[3] VAE



- **Encoder:** input $x \xrightarrow{\text{red arrow}} \mu$ and σ for posterior.
.... generate z !
- **Decoder** $z \xrightarrow{\text{red arrow}} \mu$ and σ for generative, generate $\hat{x}$

[2] VAE

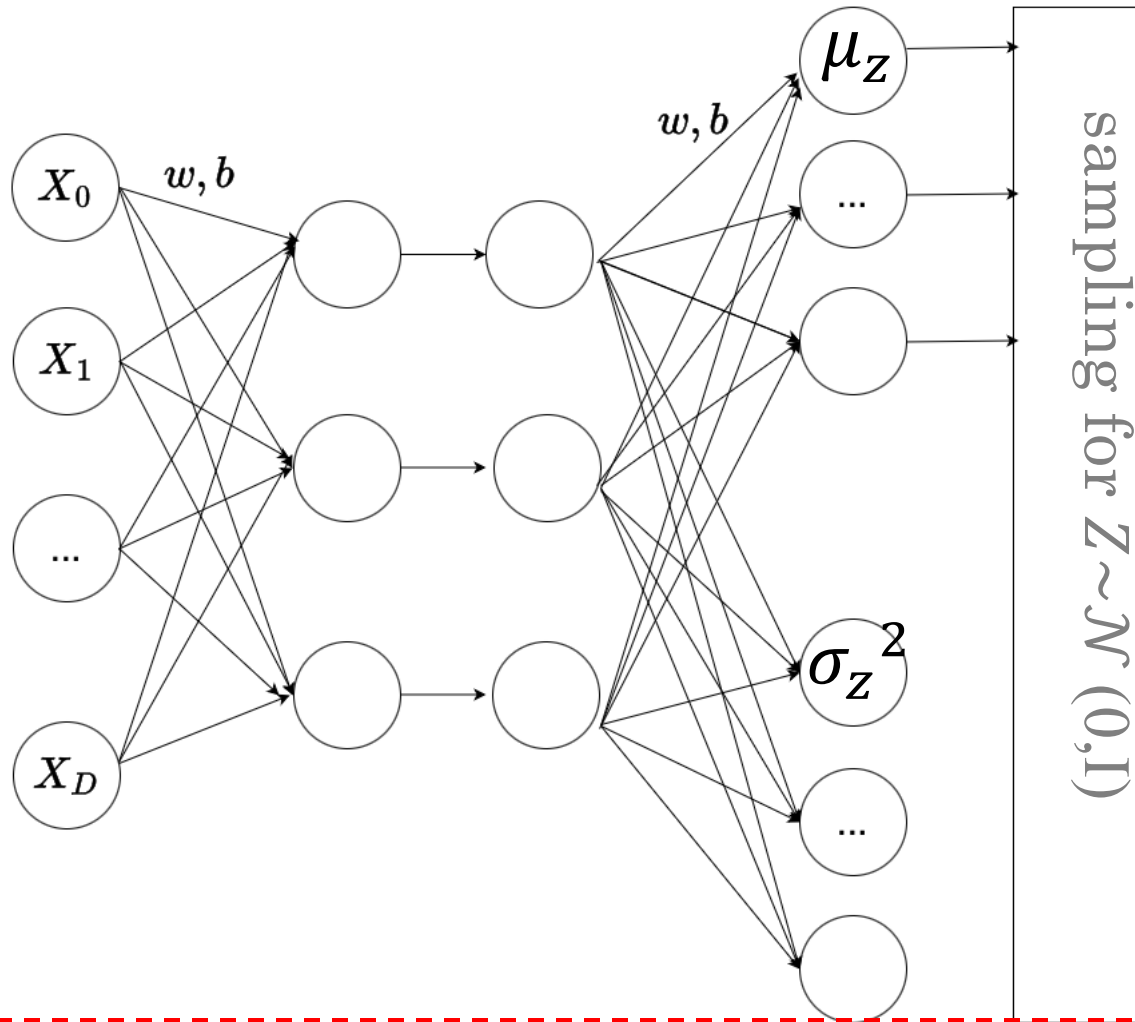


- $\mu(x), \sigma(x), \mu(z), \sigma(z)$
- the function block can be implemented **using neural nets.**

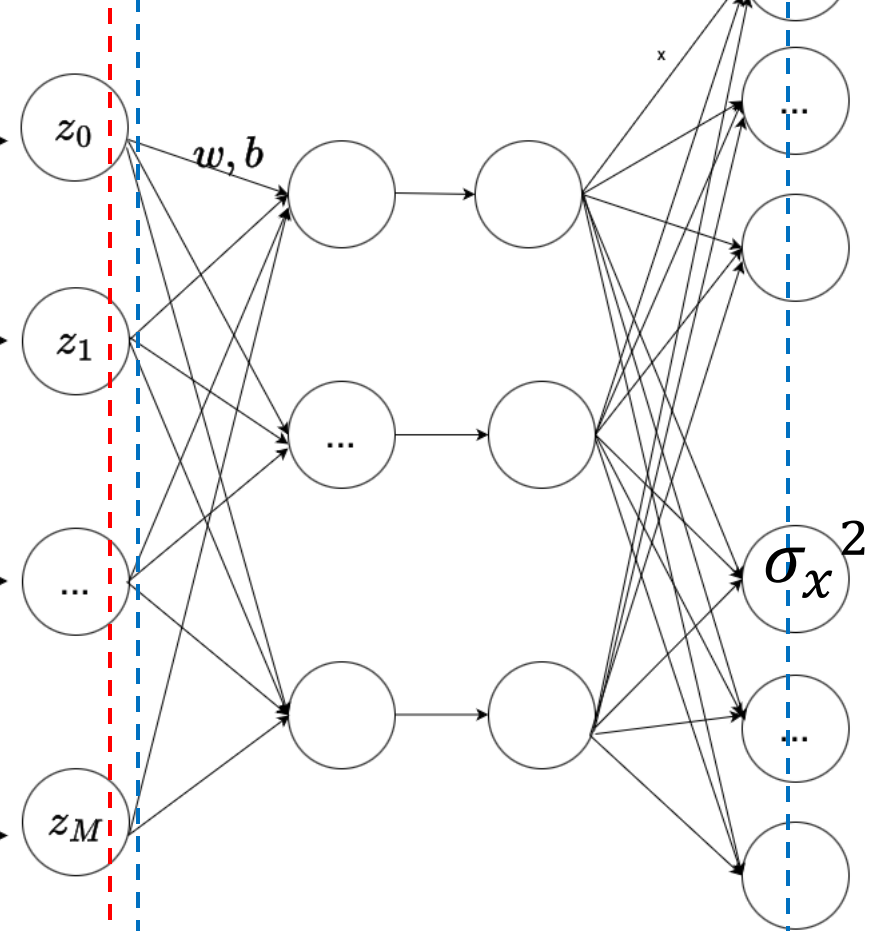
- VAE (Variational Auto-Encoder): neural net implementation
 - architecture
 - ELBO computation
 - end-to-end training, reparameterization trick

[1] VAE architecture (one example)

■ encoder : $Q(Z|X, q)$



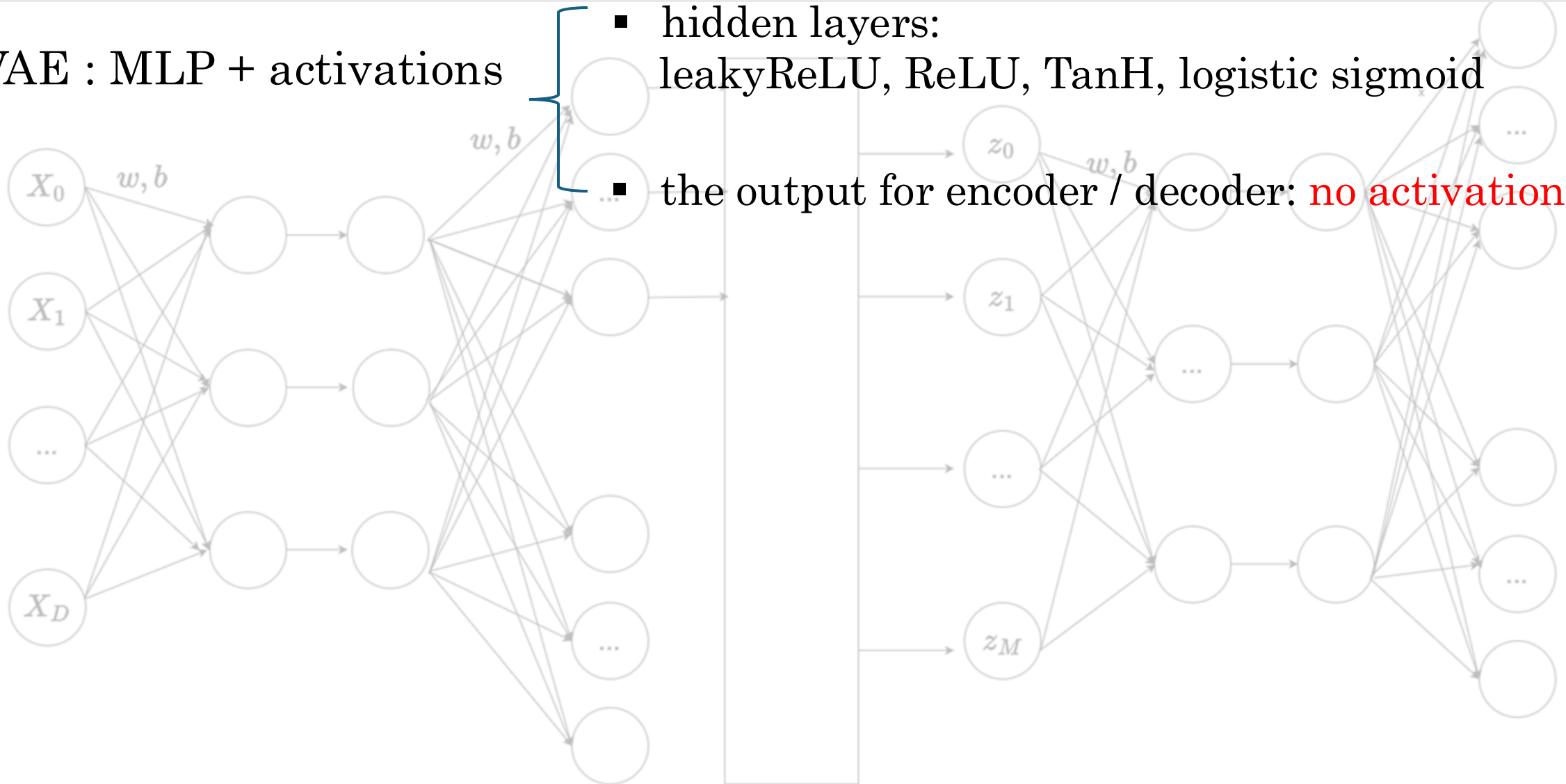
■ decoder : $P(Z|X, p)$



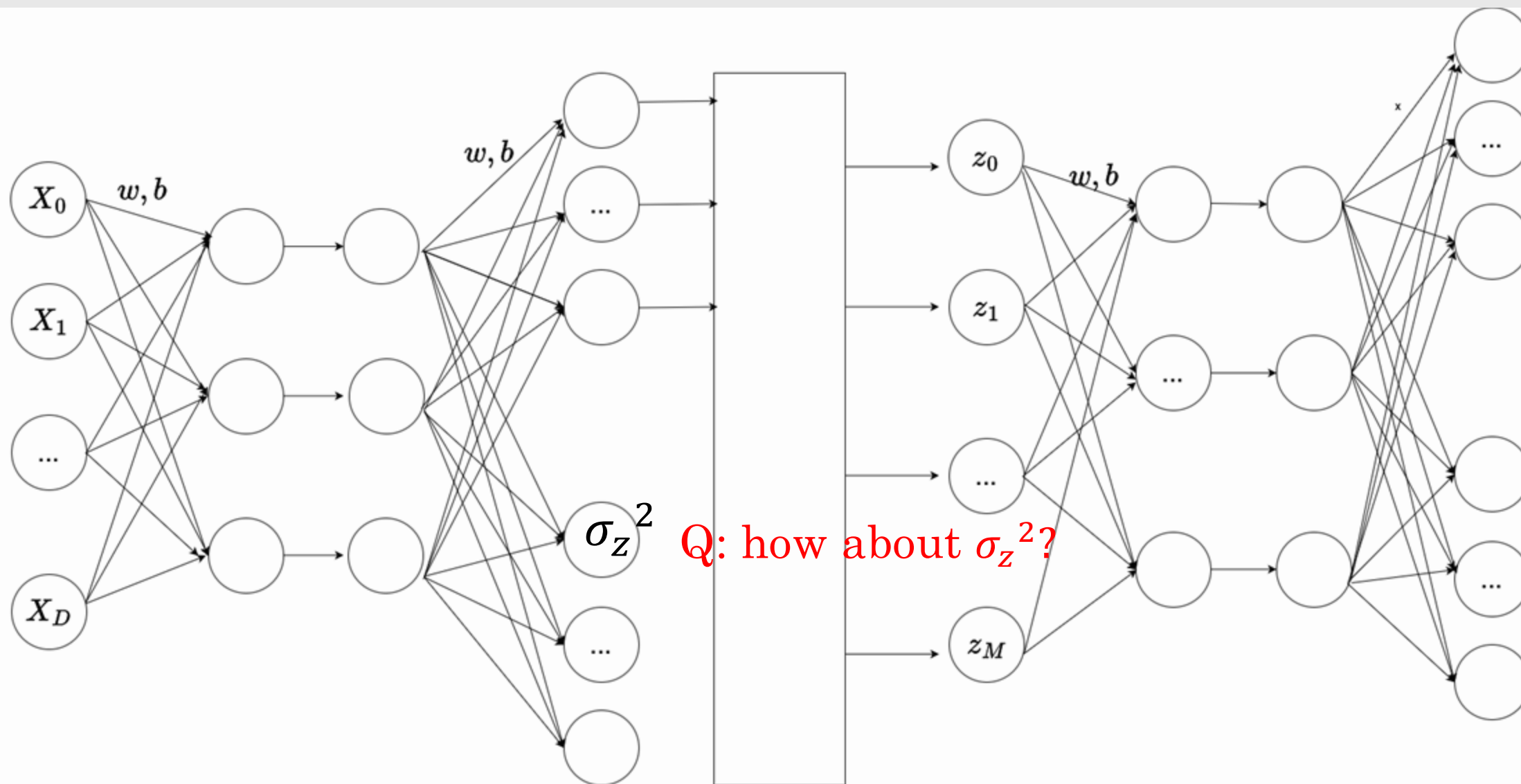
[2] VAE architecture

VAE : MLP + activations

- hidden layers:
leakyReLU, ReLU, TanH, logistic sigmoid
- the output for encoder / decoder: **no activation**

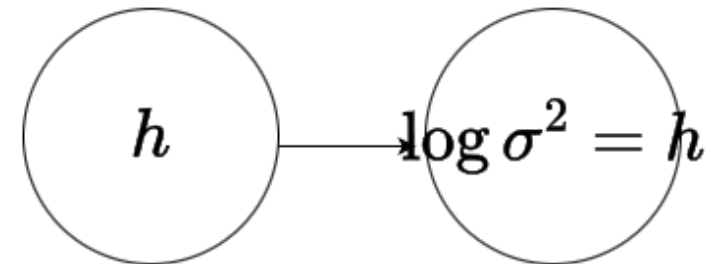
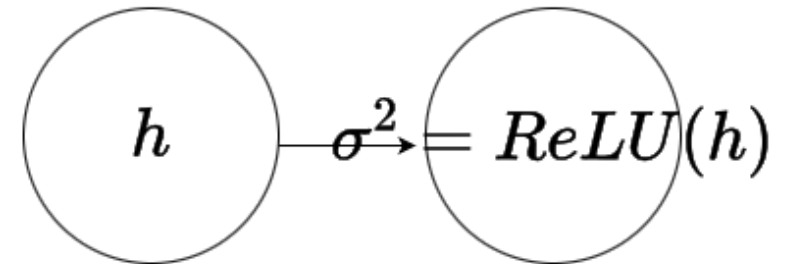
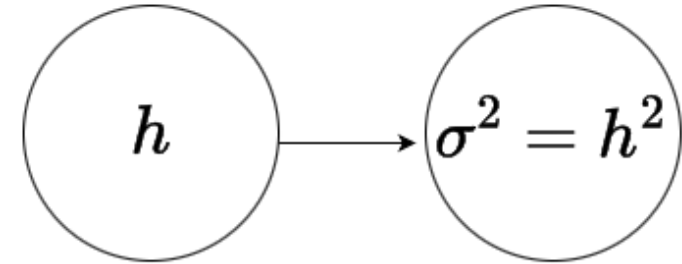


[3] VAE architecture



[4] VAE architecture

- some possible implementations for σ^2 but we generally use...



- VAE Loss: ELBO

[1] VAE Loss: ELBO

$$\begin{aligned}\log P(x_i|p) &\geq E_{Q(z|x_i,q)}\left[\log \frac{P(x_i, z|p)}{Q(z|x_i, q)}\right] \text{ [ELBO]} \\ &= E_{Q(z|x_i,q)}[\log P(x_i, z|p)] + H_{Q(z|x_i,q)}(\log Q(z|x, q)) \\ &\simeq 1/L \sum_{l=1}^L \{\log P(z_{i,l}|p) + \log P(x_i|z_{i,l}, p) - \log Q(z_{i,l}|x_i, q)\}\end{aligned}$$

[Monte Carlo Estimates of Expectations]

**Monte Carlo Estimates of Expectation

$$E_{f(X)}[Q(X)] = \int Q(x)f(x)dx$$
$$\simeq 1/L \sum_{l=1}^L Q(x_l)$$

- x_l are generated from the density $f(x)$

[2] VAE Loss (Monte Carlo Estimate)

$$\begin{aligned} & E_{Q(Z|x_i, \phi)}[P(x_i, z)] + H_{Q(Z|x_i, \phi)}(Q(Z|x_i, \phi)) \\ & \simeq 1/L \sum_{l=1}^L \{ \log p(z_{i,l}|\theta) + \log p(x_i|z_{i,l}, \theta) - \log Q(z_{i,l}|x_i, \phi) \} \\ & = 1/L \sum_{l=1}^L \left\{ -1/2 \|z_{i,l}\|^2 + \log \frac{1}{\sqrt{2\pi\sigma_{dec}^2(i, l)}} - \frac{1}{2\sigma_{dec}^2(i, l)} \|x_i - \mu_{dec}(i, l)\|^2 \right. \\ & \quad \left. - \log \frac{1}{\sqrt{2\pi\sigma_{enc}^2(i, l)}} + \frac{1}{2\sigma_{enc}^2(i)} \|z_{i,l} - \mu_{enc}(i)\|^2 \right\} + C \end{aligned}$$

- prevent having the high prob everywhere!
- reconstruction error!
- prevent collapsing!

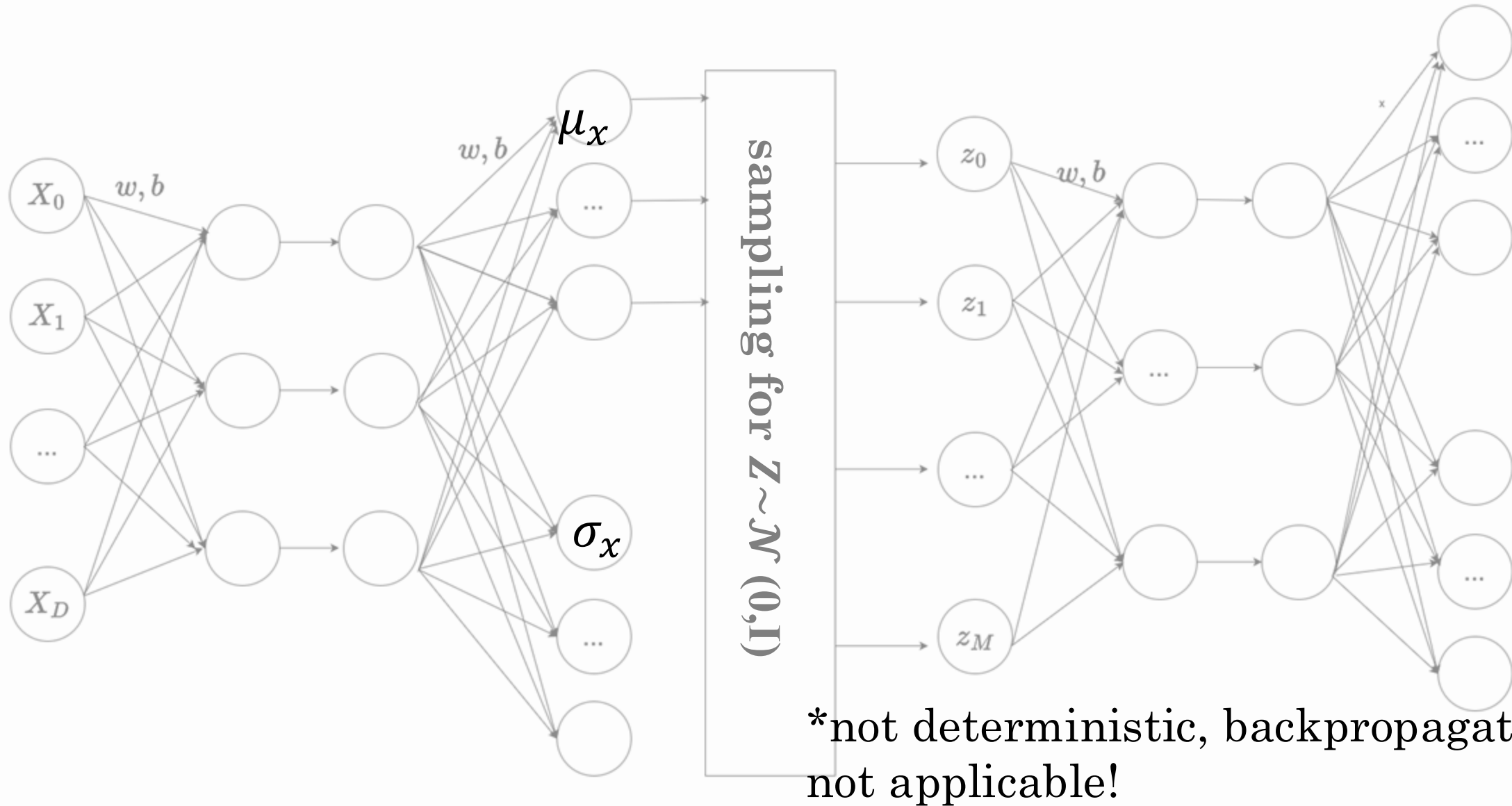
[3] VAE Loss (analytic solution)

- when using Gaussian family, the analytical solution is also possible.

Check Reference : Appendix B! <https://arxiv.org/abs/1312.6114>

- Reparameterization Trick (for end-to-end training)

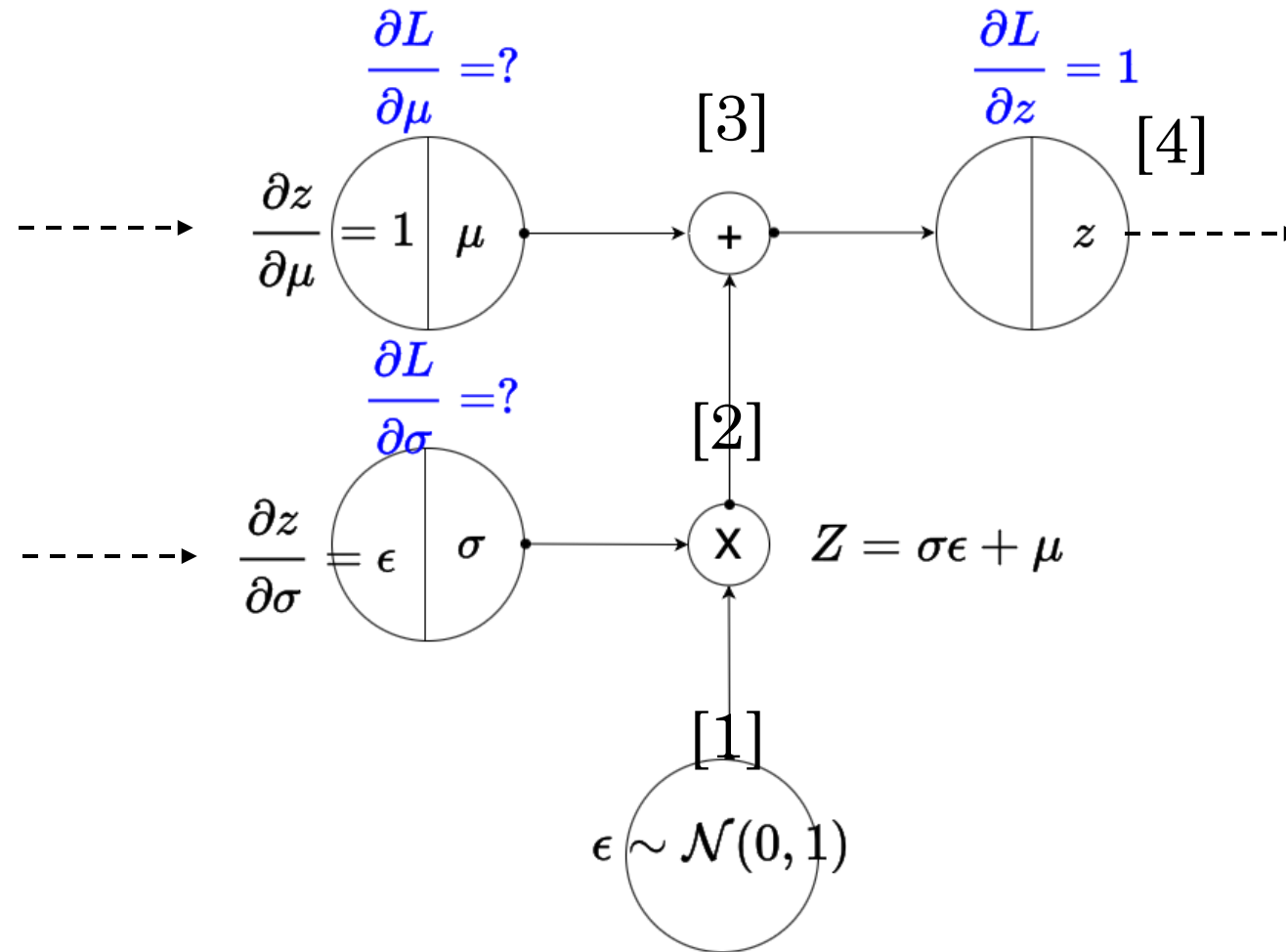
**VAE (one example architecture)



[1] Training VAE (Reparameterization Trick)

- VAE uses a special trick to generate Z sample through a deterministic function $f(\mu_x, \sigma_x, \epsilon)$
- $\epsilon \sim N(0,1)$
- $Z = f(\mu_x, \sigma_x, \epsilon) = \sigma_x \cdot \epsilon + \mu_x = N(\mu_x, \sigma_x^2)$

[2] Training VAE (Reparameterization Trick)



[2] Training VAE (End to End Learning)

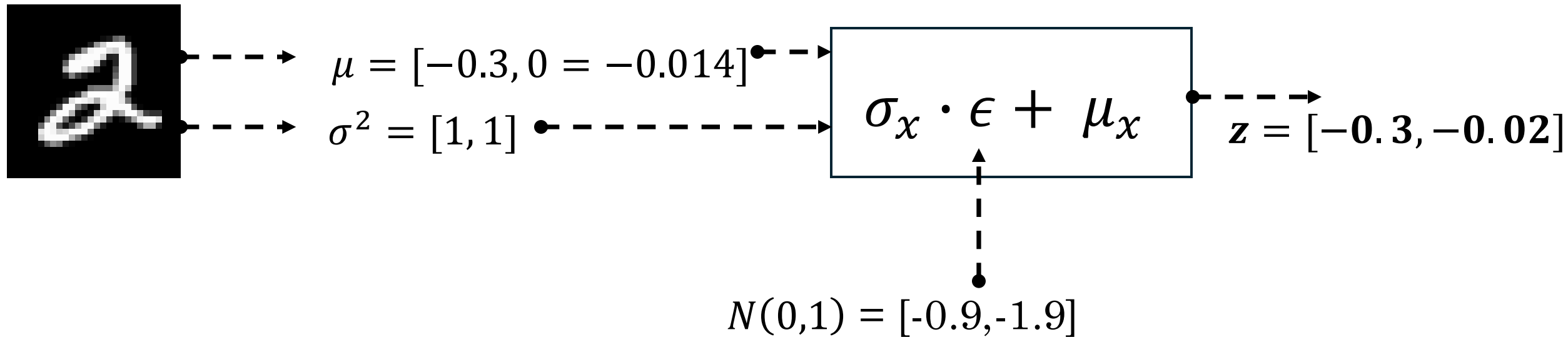
- End to End Learning:
all parts of the model from input to output together
can be updated by backpropagation loops.
-
- (1) encoder $Q(z|x, q)$ & decoder $P(z|x, p)$ in one framework
 - (2) design $Q(z|x, q)$ as a tractable family
 - (3) reparameterization trick

- VAE Experimental Results with MNIST

- Representation (Encoder)

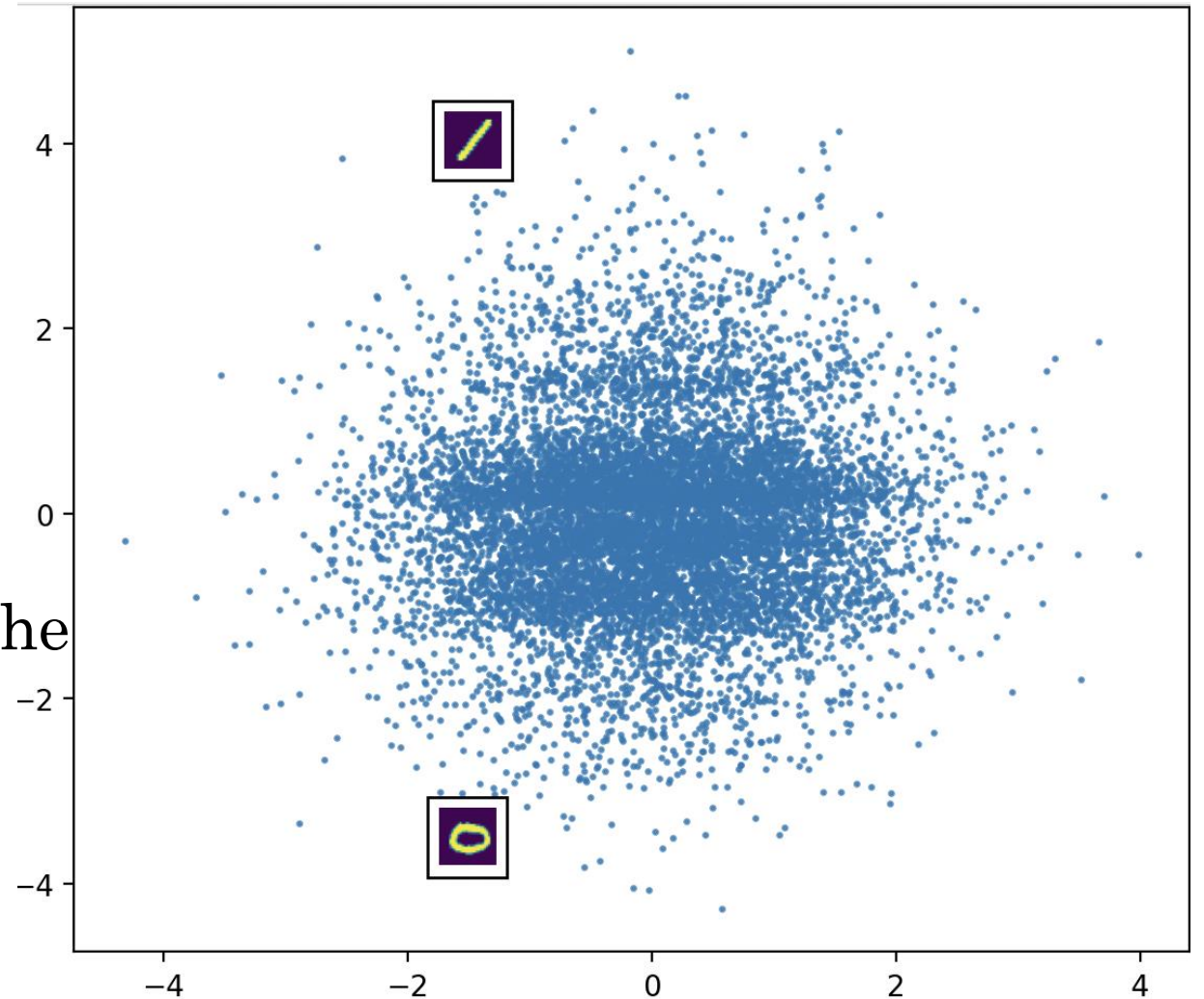
[Encoder, One Single Data Point]

- one data point (784-D)
- representation (2-D)



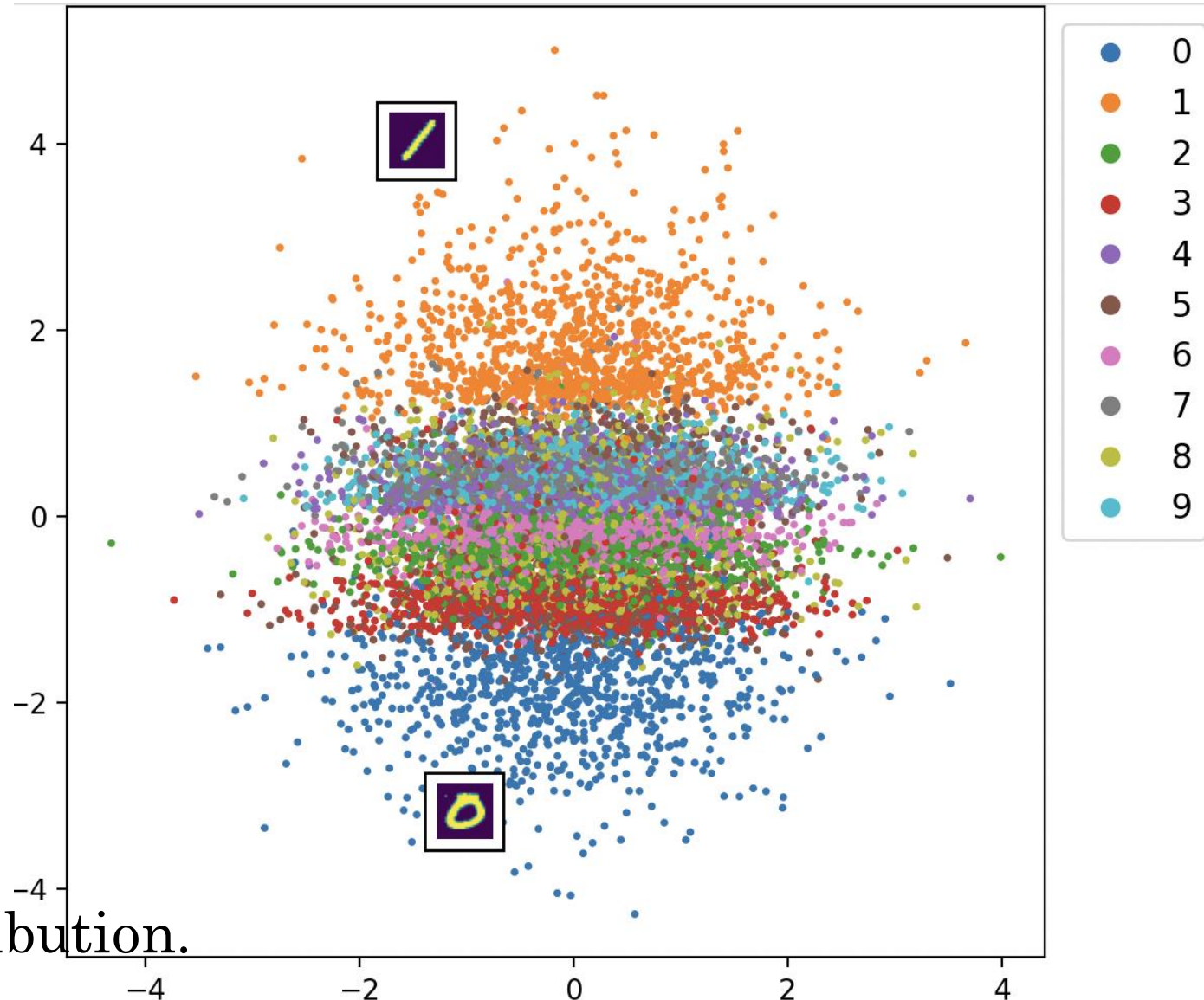
[Encoder Collective State]

- VAE assumes $Z \sim N(0, I)$
- MNIST 10,000 test samples show the **Gaussian**.



[Encoder for MNIST TEST]

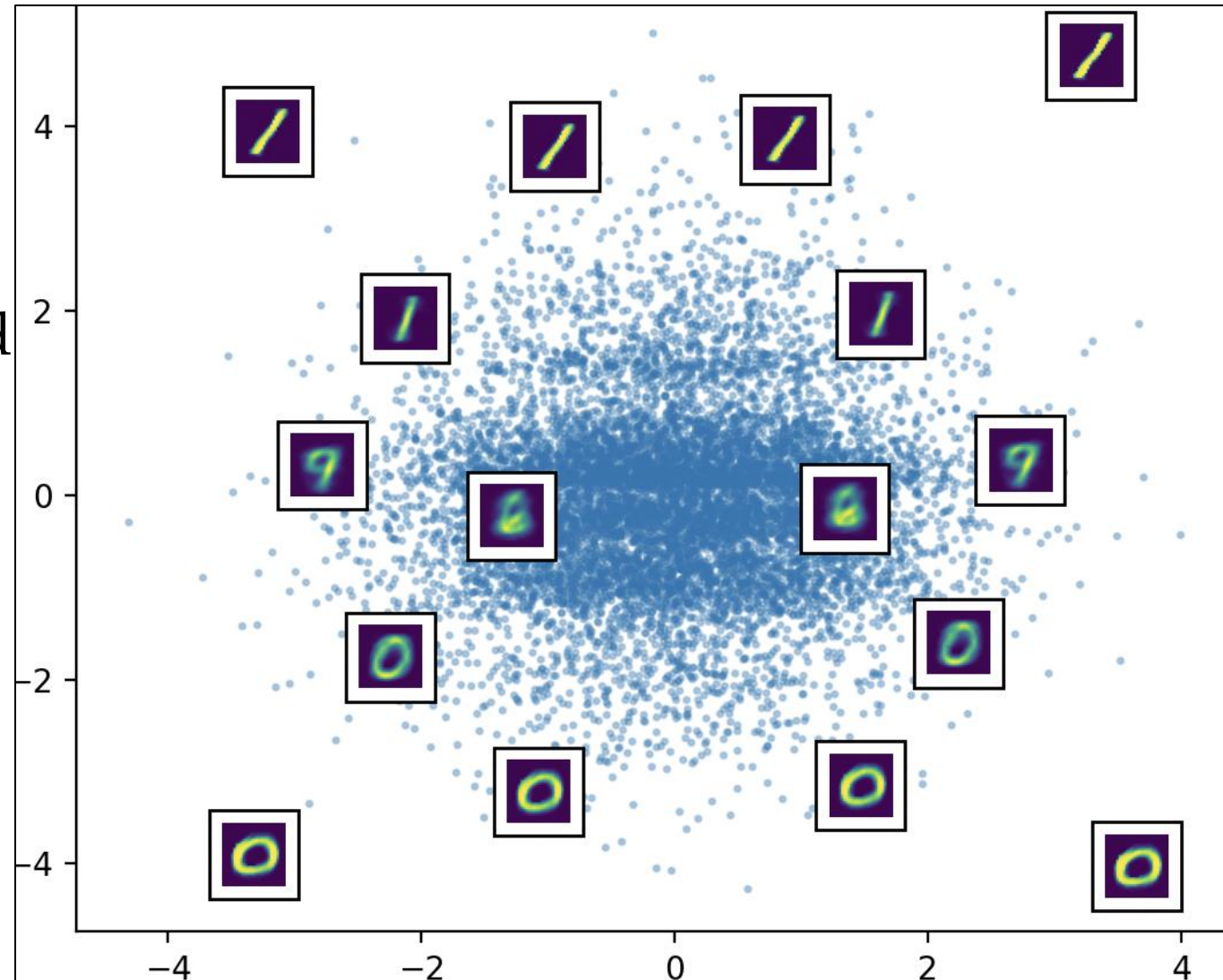
- color-coded clusters by digits
- the latent space captures the **salient feature** of the distribution.



- Generation (Decoder)

[Decoder, New Generation]

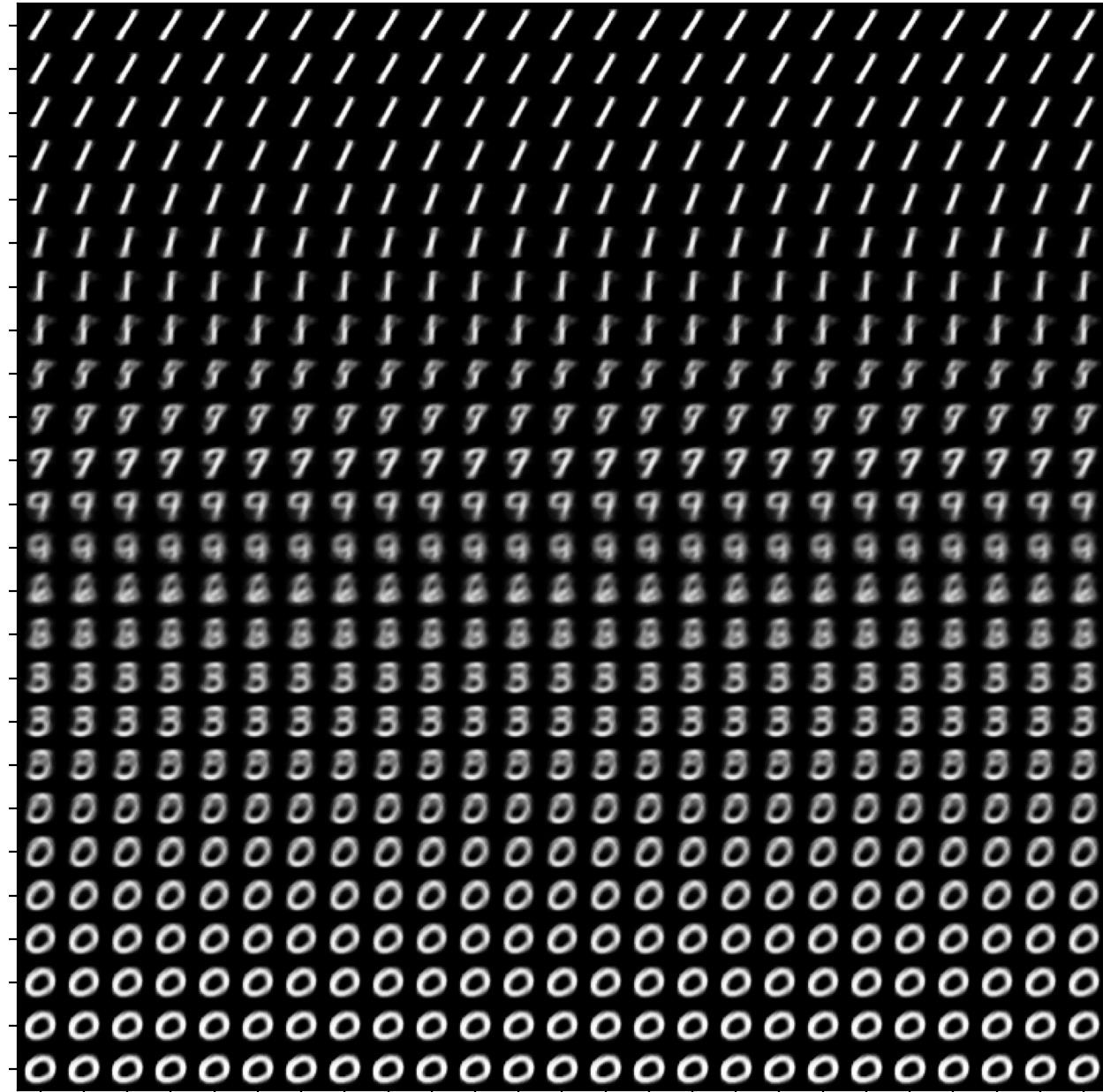
- By using the decoder, we can generate **new data** beyond the original data.



[Decoder, New Generation]

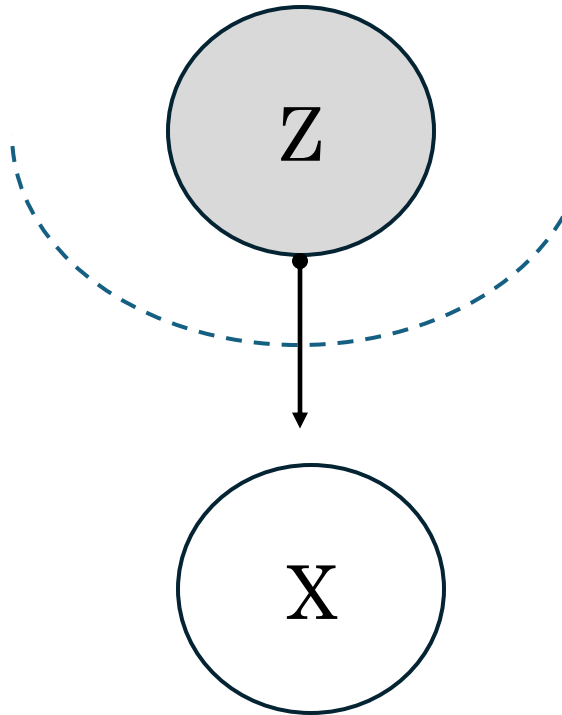
- we can see that the latent space encode **smooth and continuous** transitions between the digits.

VAE Latent Space Visualization



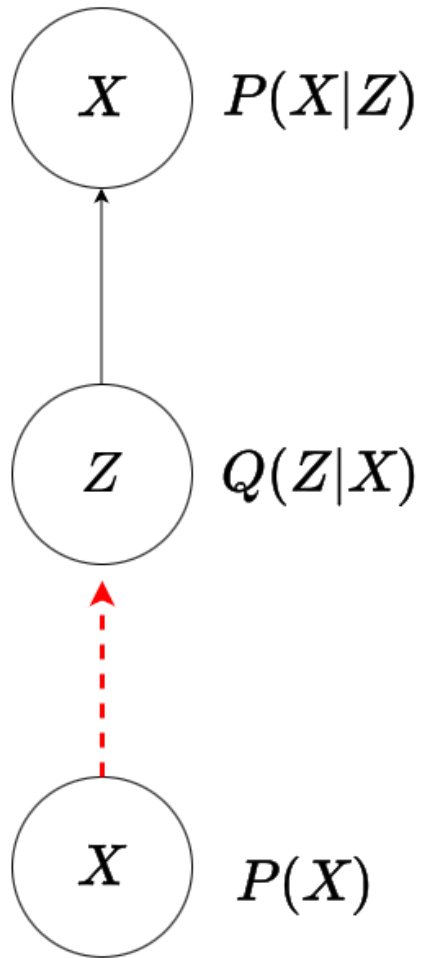
- Summary

*Summary1



- VAE is a probabilistic modeling with latent variables.

*Summary2



For intractability, $P(Z|X)$
VAE defines a new model: $Q(Z|X)$

*Summary3

- VAE optimizes ELBO aiming to approximate $P(X|p^*)$ and $P(Z|X, p^*)$

*Summary4

- VAE supports end-to-end training using the reparameterization trick.

*Summary5

- VAE learns latent space, continuous, smooth, structured, and semantically interpretable.

- VAE is

practical.

- representation space
- structured, continuous, semantic
- generative

elegant.

- architecture
- loss function
- suboptimal but enough rich to achieve the goal: representation.

simple.

- easy to train